

**TRAFFIC IMPACT ASSESSMENT FOR THE PROPOSED REZONING
AND SUBDIVISION OF ERF 11305 WALMER, WITHIN
NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE**



JULY 2026

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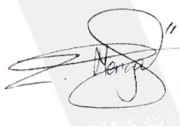
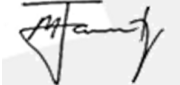
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QUALITY CONTROL DOCUMENT

CLIENT:	NELSON MANDELA BAY MUNICIPALITY
PROJECT:	TRAFFIC IMPACT ASSESSMENT FOR THE PROPOSED REZONING AND SUBDIVISION OF ERF 11305 WALMER, WITHIN NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE
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It is herewith certified that this Traffic Impact Assessment / Traffic Impact Statement has been prepared according to the requirements of the South African Traffic Impact and Site Traffic Assessment Manual.

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Traffic Impact Assessment TRAFFIC IMPACT ASSESSMENT FOR THE PROPOSED REZONING AND SUBDIVISION OF ERF 11305 WALMER, WITHIN NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE Report Summary	
Site Location	The project is in Walmer, within Nelson Mandela Bay Municipality.
Municipality	Nelson Mandela Bay Municipality.
Type of Assessment	Traffic Impact Assessment
Proposed Land use/ Development	Township Establishment
Site Size	43.73 Ha
Traffic counts date	02 June 2026
Assessment Years	Scenario 1: 2026 existing traffic volumes (Base year) Scenario 2: 2031 background traffic volumes (Design horizon year) Scenario 3: 2031 background traffic volumes + Development Scenario 4: 2036 background traffic volumes + Development (Planning horizon year)
Recommendations	The proposed development should be considered favorably from a traffic engineering point of view by the relevant authorities, given the proposed road upgrades in this document. The detailed designs of the upgrades should be undertaken by a professional engineer / technologist with suitable road design experience.
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Report Date	06 July 2026

Table of Contents

1.	KEY DEFINITIONS	6
1.1	Capacity	6
1.2	Traffic Volume	6
1.3	Volume-to-Capacity Ratio (V/C Ratio)	7
1.4	Level of Service (LOS)	7
1.5	Peak Hour Factor (PHF)	8
1.6	Base Saturation Flow	8
1.7	Analysis Horizon	9
1.8	Determination of Road Upgrading Requirements	9
1.9	Acceptable Traffic Operational Conditions	9
2.	INTRODUCTION	11
2.1	Background	11
2.2	Methodology	11
2.3	Locality and Landuse Rights	12
2.3.1	Current Landuse Rights	13
2.3.2	Proposed Landuse Rights	13
3.	SITE INVESTIGATION	14
3.1	Existing Surrounding Road Network	14
3.2	Existing Geometry	14
4.	TRAFFIC SURVEYS AND FLOW DATA	18
4.1	Turning Movement Traffic Counts	19
5.	TOWNSHIP ROAD NETWORK AND ACCESSIBILITY	32
6.	TRAFFIC DEMAND	34
6.1	Trip Generation	34
6.2	Trip Distribution and Assignments	36
6.2.1	Background	36
6.2.2	Trip Distribution and Assignment	36
6.3	Traffic Growth	36
7.	TRAFFIC IMPACT AND CAPACITY ANALYSES	38
7.1	Introduction	38
7.2	Analysis Scenarios	38
8.	PROPOSED UPGRADES	41
9.	PUBLIC TRANSPORT AND NON-MOTORISED TRANSPORT	45
9.1	Proposed Public Transport Facilities	46
9.2	Proposed Non-Motorised Transport Facilities	47
10.	CONCLUSIONS AND RECOMMENDATIONS	49
10.1	Conclusions	49
10.2	Recommendations	49
11.	REFERENCES	50

TABLE OF FIGURES

Figure 2-1: Locality Plan	12
Figure 4-1: Traffic Count Locations.....	18
Figure 4-2: Hourly Traffic Volumes (Intersection 1)	20
Figure 4-3: Hourly Traffic Volumes (Intersection 2)	22
Figure 4-4: Hourly Traffic Volumes (Intersection 3)	24
Figure 4-5: Hourly Traffic Volumes (Intersection 4)	26
Figure 4-6: Hourly Traffic Volumes (Intersection 5)	28
Figure 4-7: Hourly Traffic Volumes (Intersection 6)	30
Figure 5-1: Development Access Points	32
Figure 9-1: Modal Split	45
Figure 9-2: Pedestrian Volumes	46

LIST OF TABLES

Table 1-1: Level of Service Definitions for Intersections.....	8
Table 2-1: Landuse Summary	13
Table 3-1: Existing Intersection Geometry.....	15
Table 4-1: Summary of 12-hour vehicle traffic volumes	19
Table 4-2: Peak hour traffic volumes at Beethoven Avenue/ Road A (Intersection 1)	19
Table 4-3: Peak hour traffic volumes at Schubert Road/ Beethoven Avenue (Intersection 2)	21
Table 4-4: Peak hour traffic volumes at M18/ Road B (Intersection 3)	23
Table 4-5: Peak hour traffic volumes at M18/ Road C (Intersection 4).....	25
Table 4-6: Peak hour traffic volumes at M18/ Glendore Road (Intersection 5)	27
Table 4-7: Peak hour traffic volumes at M18/ Access to Smartstone Access (Intersection 6).....	29
Table 6-1: Weekday AM Trip Generation	35
Table 6-2: Weekday PM Trip Generation	35
Table 6-3: Weekend Trip Generation.....	35
Table 7-2: Capacity Analysis Results	39
Table 8-1: Proposed Upgrades	41

LIST OF ANNEXURES

Annexure A: Township Layout
Annexure B: Traffic Volumes
Annexure C: Scenario 1 Capacity Analysis Results
Annexure D: Scenario 2 Capacity Analysis Results
Annexure E: Scenario 3 Capacity Analysis Results
Annexure F: Scenario 4 Capacity Analysis Results

LIST OF ACRONYMS

COTO	Committee of Transport Officials
LOS	Level of Service
TIA	Traffic Impact Assessment
TMH	Technical Methods for Highways
v/c	Volume to Capacity ratio
Veh/h	Vehicles per hour
s/veh	Seconds per vehicle
PCU/h	Passenger Car Units per hour
PT	Public Transport
NMT	Non-Motorized Transport
HCM	Highway Capacity Manual
PHF	Peak Hour Factor
IDP	Integrated Development Plan
TRH	Technical Recommendations for Highways

TRAFFIC IMPACT ASSESSMENT FOR THE PROPOSED REZONING AND SUBDIVISION OF ERF 11305 WALMER, WITHIN NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE

1. KEY DEFINITIONS

This chapter presents the key terminology and technical definitions applied in the analysis and assessment undertaken in this study. The definitions are primarily based on recognized traffic engineering references, including the Highway Capacity Manual (HCM) and relevant South African guidelines issued by the Committee of Transport Officials (COTO).

1.1 Capacity

Capacity is defined as the maximum hourly rate at which vehicles can reasonably be expected to traverse a lane or roadway segment during a given period under prevailing roadway, traffic, and traffic control conditions.

Capacity is influenced by factors such as:

- Roadway geometry
- Traffic composition
- Traffic control devices
- Driver behaviour
- Environmental conditions

Capacity is typically expressed in vehicles per hour (veh/h) or passenger car units per hour (PCU/h).

1.2 Traffic Volume

Traffic volume refers to the actual number of vehicles passing a specific point or section of roadway during a defined period, usually expressed as an hourly rate.

Traffic volume is commonly expressed as:

- **Vehicles per hour (veh/h)**

Traffic volumes are used to assess roadway demand and to evaluate the operational performance of transport infrastructure.

1.3 Volume-to-Capacity Ratio (V/C Ratio)

$V/C = \text{Traffic Volume} / \text{Capacity}$

The V/C ratio is a key indicator of traffic performance and is used to determine the degree of congestion and operational efficiency of a roadway or intersection.

In general:

- **V/C < 1.0** indicates that capacity is sufficient to accommodate traffic demand
- **V/C = 1.0** indicates that demand is equal to capacity
- **V/C > 1.0** indicates that demand exceeds capacity, resulting in congestion

1.4 Level of Service (LOS)

Level of Service (LOS) is a qualitative measure used to describe the operational performance of a roadway or intersection from the perspective of road users. It reflects the degree of congestion, delay, and driver comfort associated with traffic conditions.

Level of Service is typically defined in terms of control delay, which represents the additional travel time experienced by vehicles due to traffic control measures such as signals, stop signs, or congestion.

Delay is an indicator of:

- Driver discomfort and frustration
- Increased travel time
- Fuel consumption
- Reduced operational efficiency

Table 1-1: Level of Service Definitions for Intersections

Level of Service	Signalised Intersections (Control Delay per Vehicle, s/veh)	Unsignalised Intersections (Control Delay per Vehicle, s/veh)
A	< 10	< 10
B	10 – 20	10 – 15
C	20 – 35	15 – 25
D	35 – 55	25 – 35
E	55 – 80	35 – 50
F	> 80	> 50

1.5 Peak Hour Factor (PHF)

The Peak Hour Factor (PHF) is a measure used to account for variations in traffic flow within the peak hour. It represents the relationship between the total hourly traffic volume and the highest rate of flow observed during a shorter time interval within that hour.

The Peak Hour Factor is defined as:

$$\text{PHF} = \text{Total Hourly Volume} / (4 \times \text{Highest 15-minute Volume})$$

The PHF is used to:

- Adjust traffic volumes for peak flow conditions
- Assess traffic demand variability
- Improve the accuracy of capacity and operational analyses

Peak Hour Factors were calculated for the intersections assessed in this study and applied to the observed traffic volumes where required.

1.6 Base Saturation Flow

Base Saturation Flow is a key traffic performance measure representing the maximum rate at which vehicles can pass through an intersection approach lane under ideal conditions during the green phase of a signal cycle.

The base saturation flow rates adopted for this study are presented below.

Base Saturation Flow Rates

Approach Type	Base Saturation Flow Rate (PCU/hr/lane)
Through lanes	2050
Right-turn, left-turn and shared lanes	1600
Slip lanes (give-way or signalised)	1800

1.7 Analysis Horizon

An analysis horizon of 2031 and 2036 was adopted in this study, in accordance with the COTO TMH 16: South African Traffic Impact and Site Traffic Assessment Manual (Version 1.0, August 2012). The selected horizon years reflects the anticipated operational conditions during the planning and implementation period associated with the proposed infrastructure works and potential access changes.

1.8 Determination of Road Upgrading Requirements

The determination of road upgrading requirements associated with operational, or infrastructure changes is guided by the: COTO TMH 15: South African Engineering Service Contribution Manual for Municipal Road Infrastructure (Version 1.0, September 2012)

In accordance with this guideline:

- Internal engineering services are the responsibility of the project proponent or facility operator
- External engineering services are typically the responsibility of the municipality
- Contributions toward external infrastructure upgrades may be required where operational changes impact the surrounding road network

1.9 Acceptable Traffic Operational Conditions

Traffic operations are generally considered acceptable when the following conditions are satisfied during the peak 15-minute period of the assessment hour:

- a) The Volume-to-Capacity Ratio (V/C) does not exceed 1.0, indicating that demand does not exceed the capacity of the facility; and
- b) The Level of Service (LOS) meets the minimum acceptable operational standard for the relevant road classification.

Right-turn movements at signalised intersections may operate at Level of Service E, provided that adequate storage capacity is available to accommodate expected queue lengths.

Acceptable Levels of Service for Roadway Operations

Area / Road Classification	Normal Operating Conditions	Abnormal Operating Conditions
Urban Roads	LOS D	LOS E
Rural Roads (Class 3–5)	LOS C	LOS D
Rural Roads (Class 1–2)	LOS B	LOS C

Definitions of normal and abnormal operating conditions are provided in the TRH 17: South African Trip Data Manual (COTO 2012).

2. INTRODUCTION

2.1 Background

Gwaca Consulting cc was appointed by the Nelson Mandela Bay Municipality to conduct a Traffic Impact Assessment (TIA) in support a proposed rezoning and subdivision of Erf 11305 Walmer located within the Nelson Mandela Bay Municipality, Eastern Cape Province. The development will comprise of residential I, residential IV, business I, and institutional land uses. Most of the land uses except for the residential 1 & 4 use will serve the local community and will not have an impact on the external road network.

This report presents the findings of the Traffic Impact Assessment and provides an evaluation of the anticipated traffic implications arising from the proposed development. It further outlines recommended mitigation measures and identifies any road infrastructure upgrades that may be required to accommodate the projected increase in traffic demand.

The Township Layout Plan is included as **Annexure A** for reference.

2.2 Methodology

The TIA for the proposed rezoning and subdivision was undertaken based on the total anticipated traffic demand associated with Erf 11305 Walmer. A five-year design horizon, and a ten-year planning horizon, was adopted to assess whether the existing transport infrastructure serving the site and the surrounding road network will be capable of accommodating the additional traffic generated by the proposed development.

The TIA was compiled in accordance with the following methodology:

- Development-related information was reviewed to confirm the proposed land use, establish the site location, and inform the scope of site investigations.
- Key road links and critical intersections serving as primary access routes to and from the study area were assessed to determine existing road geometry, traffic safety conditions, and base-year traffic volumes.
- Traffic generation associated with the proposed development was estimated based on the applicable land use rights and development size.
- Trip distribution and assignment for the development-generated traffic was derived from observed traffic flow patterns within the surrounding road network.
- Background traffic growth was applied over a five-year and a ten-year horizon using an annual growth rate of 2%. Existing intersection operating conditions were analysed using the SIDRA intersection analysis software. Development-generated traffic was then

assigned to the network and assessed under both base-year and future background traffic scenarios.

The existing public transport infrastructure within the study area was also evaluated, and where necessary, recommendations were made for upgrades or improvements to accommodate the proposed development

2.3 Locality and Landuse Rights

The proposed site for the township establishment is located within the jurisdiction of the Nelson Mandela Bay Municipality, on Erf 11305 Walmer. The location of the proposed development is illustrated in **Figure 2-1**.



Figure 2-1: Locality Plan

2.3.1 Current Landuse Rights

The study area is characterised by a mix of land uses, including greenfields, residential developments, places of public worship, schools, hospitals, shopping centres, railway and primary and secondary road networks.

2.3.2 Proposed Landuse Rights

The proposed development site measures approximately 43.73 hectares. The proposed land use development is intended to support the following land use typologies:

Table 2-1: Landuse Summary

Township	Zoning	LANDUSE	USES PERMITTED (LANDUSE CODE/ THM 17)	EXTENT	UNITS
Erf 11305 Walmer	Residential I	Single Dwelling Units (Free Basic Housing)	210 Single Dwelling Units	509	Units
	Residential I	Single Residential Housing (Gap Housing)	210 Single Dwelling Units	143	
	Residential IV	High Density Residential	220 Apartments and Flats	300	Units
	Business I	Shops	820 Shopping Centre	1.764	Ha
	Institutional I	Public Secondary School		1500	Students
	Open Space Zone I	Open Space Zone	-	12.876	Ha
	Transport Zone	Streets	-	6.613	Ha

The business I land use planned within the development comprises tuckshops intended solely to serve the internal needs of the development's residents. Similarly, the Institutional land use was considered subservient to the residential land use and is assumed to generate only internal trips within the development. As such, these facilities are not expected to generate external trips. Consequently, the business I and the institutional I related trips have been excluded from the trip generation calculations.

3. SITE INVESTIGATION

A site visit was conducted on the 2nd of June 2026 to obtain first-hand knowledge of the existing transportation conditions and to identify potential transportation impacts that may arise from the proposed development. During the site visit, the site and the surrounding road network were assessed to understand the existing traffic patterns, pedestrian and bicycle facilities, and transit services.

The site visit provided an opportunity to observe the site's accessibility and connectivity, such as the number of access points and the level of service of intersections. It also allowed for the identification of any potential traffic bottlenecks, sight distance issues, or pedestrian safety concerns that may need to be addressed in the TIA report. The findings of the site visit are outlined in this report.

3.1 Existing Surrounding Road Network

The road network surrounding the proposed development provides both local and regional connectivity and comprises a combination of arterial, collector and local access roads. The functional characteristics of the surrounding road network are discussed below:

- **Victoria Drive (M18):** Victoria Drive functions as a Class 2 arterial route, providing an important east–west connection between Marine Drive and Buffelsfontein Road (M9). The route accommodates through traffic while also serving adjacent residential neighbourhoods and recreational land uses within the study area. Access to the southern portion of the development will be obtained from this road.
- **Glendore Road:** Glendore Road is classified as a Class 4 collector road. Its primary function is to collect and distribute traffic between the surrounding residential areas and the higher-order road network, including Victoria Drive and adjacent local streets.
- **Beethoven Avenue:** Beethoven Avenue functions as a Class 5 local access street, providing direct access to adjoining residential properties. The road also serves as a feeder to the collector road network. The second access to the northern portion of the proposed development is planned to connect to this road.
- **Schubert Road:** Schubert Road is also classified as a Class 5 local access street. It primarily provides access to adjacent residential properties and accommodates low traffic volumes typical of local residential streets.
- **Buffelsfontein Road (M9):** Buffelsfontein Road is a Class 1 arterial route and forms one of the principal north–south corridors within Nelson Mandela Bay. The route carries substantial traffic volumes between Walmer, Walmer Heights, Fairview and the northern suburbs and forms a key component of the metropolitan strategic road network.

3.2 Existing Geometry

The existing intersection and geometry control for the external intersections to be analysed can be seen in the table below.

Table 3-1: Existing Intersection Geometry

Nr	Intersection Name	Existing Layout	Description
1	Beethoven Avenue/ Road A		The intersection of Beethoven Avenue/ Road A currently operates under priority control, with a single traffic lane provided in each direction on all approaches. A site inspection confirmed that the existing road markings are worn and lack adequate visibility, reducing their effectiveness in guiding motorists. The pavement surface was observed to be in fair condition, with no major defects affecting traffic operations. The available sight distances at the intersection are considered adequate and comply with the operational requirements for safe vehicle manoeuvring and decision-making.
2	Schubert Road/ Beethoven Avenue		The intersection of Schubert Road/ Beethoven Avenue currently operates under priority control, with a single traffic lane provided in each direction on all approaches. A site inspection confirmed that the existing road markings are faded and offer limited visibility, reducing their effectiveness in guiding motorists. The pavement is generally in fair condition, with no significant defects observed that would adversely affect traffic operations. The available sight distance at the intersection is considered adequate and satisfies the requirements for safe vehicle manoeuvring and driver decision-making.

Nr	Intersection Name	Existing Layout	Description
3	M18/ Road B		The intersection of M18/Road B currently operates under priority control, with a single traffic lane provided in each direction on all approaches. A site inspection confirmed that the existing road markings are worn and exhibit poor visibility, reducing their effectiveness in guiding motorists. The pavement surface is generally in fair condition, with no significant defects observed that would adversely affect traffic operations. In addition, the available sight distance at the intersection is considered adequate and meets the operational requirements for safe vehicle manoeuvring and driver decision-making.
4	M18/ Road C		The intersection of M18/ Road C currently operates under priority control, with a single traffic lane provided in each direction on all approaches. A site inspection confirmed that the existing road markings are worn and lack adequate visibility, reducing their effectiveness in guiding motorists. The pavement surface was observed to be in fair condition, with no major defects affecting traffic operations. The available sight distances at the intersection are considered adequate and comply with the operational requirements for safe vehicle manoeuvring and decision-making.

Nr	Intersection Name	Existing Layout	Description
5	M18/ Glendore Road		The intersection of M18/ Glendore Road currently operates under priority control, with a single traffic lane provided in each direction on all approaches. A site inspection confirmed that the existing road markings are worn and lack adequate visibility, reducing their effectiveness in guiding motorists. The pavement surface was observed to be in fair condition, with no major defects affecting traffic operations. The available sight distances at the intersection are considered adequate and comply with the operational requirements for safe vehicle manoeuvring and decision-making.
6	M18/ Smartstone Access		This T-junction, comprising three approaches, currently operates under priority control. Each approach accommodates a single traffic lane in each direction, while the northern leg remains unpaved. The intersection functions as an informal access point.

4. TRAFFIC SURVEYS AND FLOW DATA

Gwaca Consulting conducted traffic surveys on the 2nd of June 2026. Traffic (pedestrian and vehicle) counts were conducted at the locations as shown in the figure below. The traffic counts that were conducted are as follows:

- Twelve-hour turning movement traffic counts (06h00 to 18h00)
- Twelve-hour classified traffic counts (06h00 to 18h00)



Figure 4-1: Traffic Count Locations

4.1 Turning Movement Traffic Counts

The 12-hour turning movement counts data is summarized in **Table 4-1**.

Table 4-1: Summary of 12-hour vehicle traffic volumes

Point	Location	Intersection 12hr Volumes				
		Cars	Taxis	Buses	Trucks	Pedestrians
P1	Beethoven Avenue/ Road A	230	0	0	6	9
P2	Beethoven Avenue/ Schubert Road	964	0	9	14	9
P3	M18/ Road B	3324	211	16	624	126
P4	M18/ Road C	2441	132	14	568	139
P5	M18/ Glendore Road	3236	130	16	258	160
P6	M18/ Access to SmartStone Access	2705	155	16	565	138

It can be noted on **Table 4-1** that the 12-hour intersection volumes for private cars are 230, 964, 3324, 2441, and 3236 and 2705 for Stations P1, P2, P3, P4, P5 and P6 respectively. The 12-hour volumes for Taxis are 0, 0, 211, 132, 130 and 155 for Stations P1, P2, P3, P4, P5 and P6 respectively. The 12-hour volumes for Buses are 0, 9, 16, 14, 16 and 16 for Stations P1, P2, P3, P4, P5 and P6 respectively. The 12-hour volumes for Trucks are 6, 14, 624, 568, 258 and 565 for Stations P1, P2, P3, P4, P5 and P6 respectively. The 12-hour volumes for Pedestrians are 9, 9, 126, 139, 160 and 138 for Stations P1, P2, P3, P4, P5 and P6 respectively.

The morning and afternoon peak hour traffic volumes for 2025 are shown in the tables and graphs below for Intersections 1,2, 3, 4, 5 and 6.

Table 4-2: Peak hour traffic volumes at Beethoven Avenue/ Road A (Intersection 1)

APPROACH	Morning Peak Volumes (06h30 to 07h30)			
	Left	Through	Right	TOTAL
Beethoven Avenue (Eastbound)	-	2	2	4
Beethoven Avenue (Westbound)	0	19	-	19
Road A (Northbound)	2	-	1	3
APPROACH	Afternoon Peak Volumes (16h30 to 17h30)			
	Left	Through	Right	TOTAL
Beethoven Avenue (Eastbound)	-	11	3	14
Beethoven Avenue (Westbound)	0	7	-	7
Road A (Northbound)	4	-	0	4

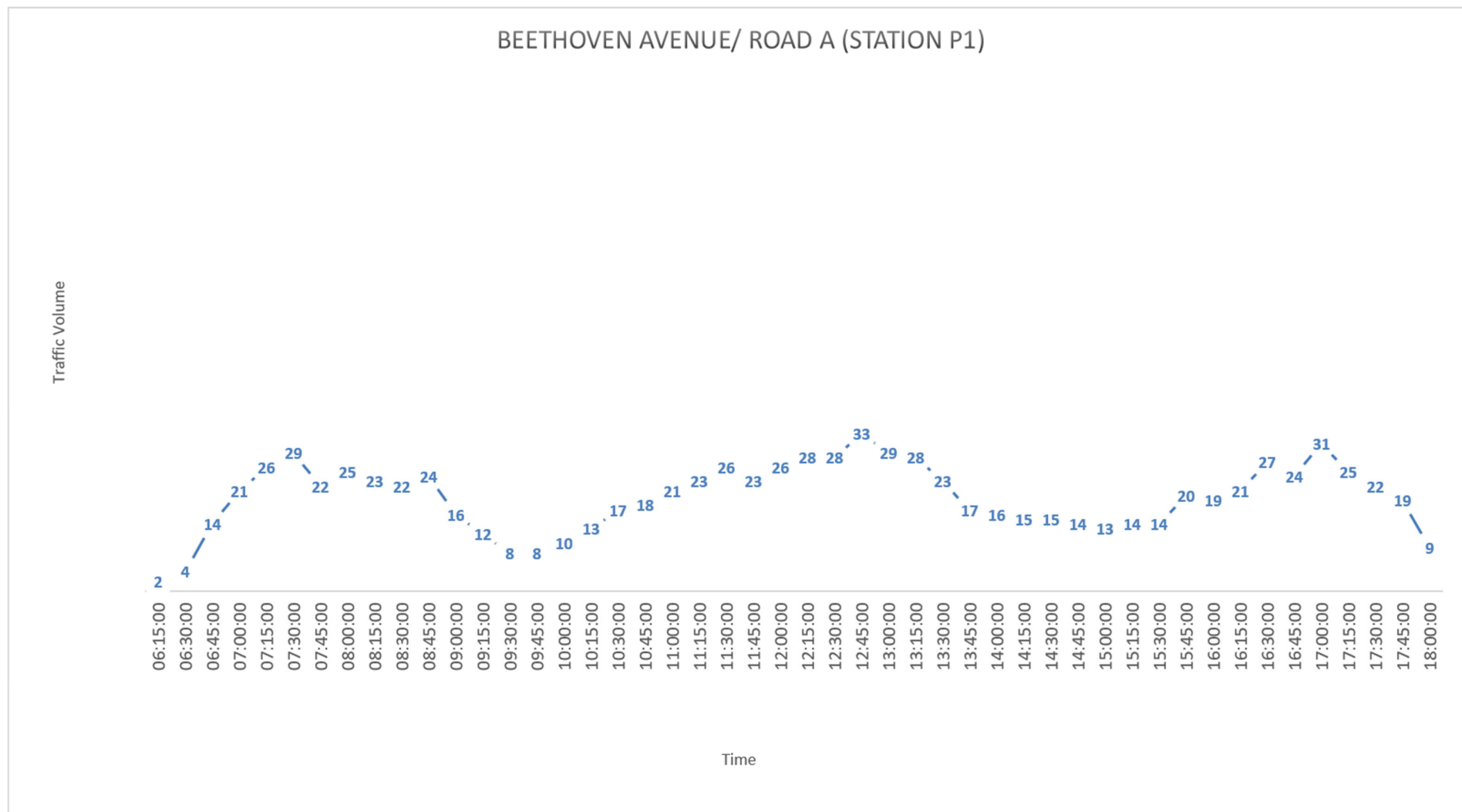


Figure 4-2: Hourly Traffic Volumes (Intersection 1)

It can be noted in **Figure 4-2** that for the Beethoven Avenue/ Road A Intersection (intersection 1), the morning peak hour is from 06h30 to 07h30 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 29 and 31 vehicles per hour for the morning and afternoon peaks respectively.

Table 4-3: Peak hour traffic volumes at Schubert Road/ Beethoven Avenue (Intersection 2)

APPROACH		Morning Peak Volumes (06h30 to 07h30)		
		Left	Through	Right
Schubert Road (Southbound)		9	-	20
Beethoven Avenue (Westbound)		-	6	29
Beethoven Avenue (Eastbound)		45	0	-
APPROACH		Afternoon Peak Volumes (16h30 to 17h30)		
Schubert Road (Southbound)		55	-	20
Beethoven Avenue (Westbound)		22	2	-
Beethoven Avenue (Eastbound)		-	3	14

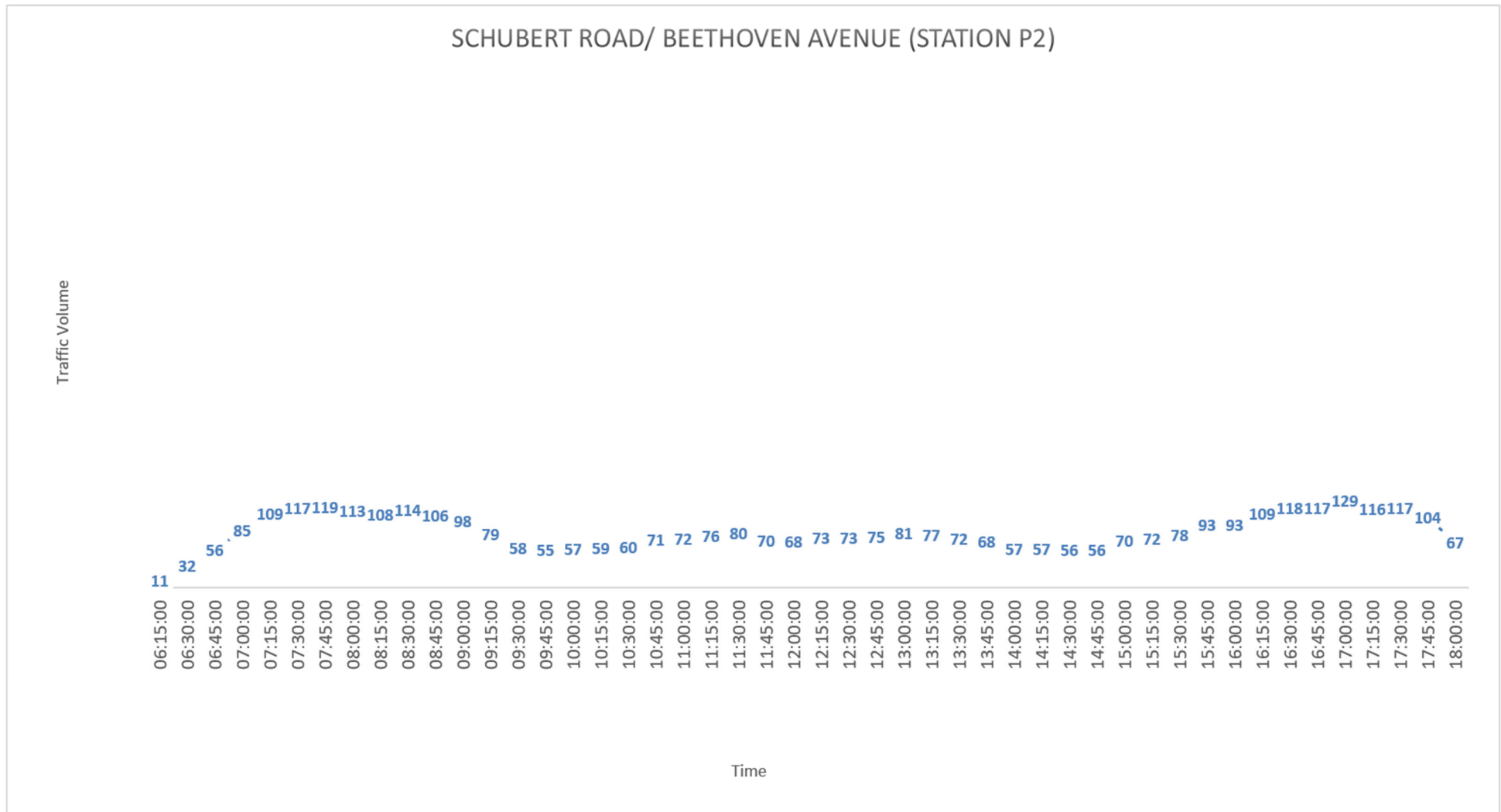


Figure 4-3: Hourly Traffic Volumes (Intersection 2)

It can be noted in **Figure 4-3** that for the Schubert Road/ Beethoven Avenue intersection (Intersection 2), the morning peak hour is from 06h30 to 07h30 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 117 and 129 vehicles per hour for the morning and afternoon peaks respectively.

Table 4-4: Peak hour traffic volumes at M18/ Road B (Intersection 3)

APPROACH	Morning Peak Volumes (07h00 to 08h00)			
	Left	Through	Right	TOTAL
Road B (Northbound)	21	-	23	44
M18 (Westbound)	26	130	-	156
M18 (Eastbound)	-	410	11	421
APPROACH	Afternoon Peak Volumes (16h30 to 17h30)			
	Left	Through	Right	TOTAL
Road B (Northbound)	7	-	31	38
M18 (Westbound)	39	252	-	291
M18 (Eastbound)	-	141	6	147

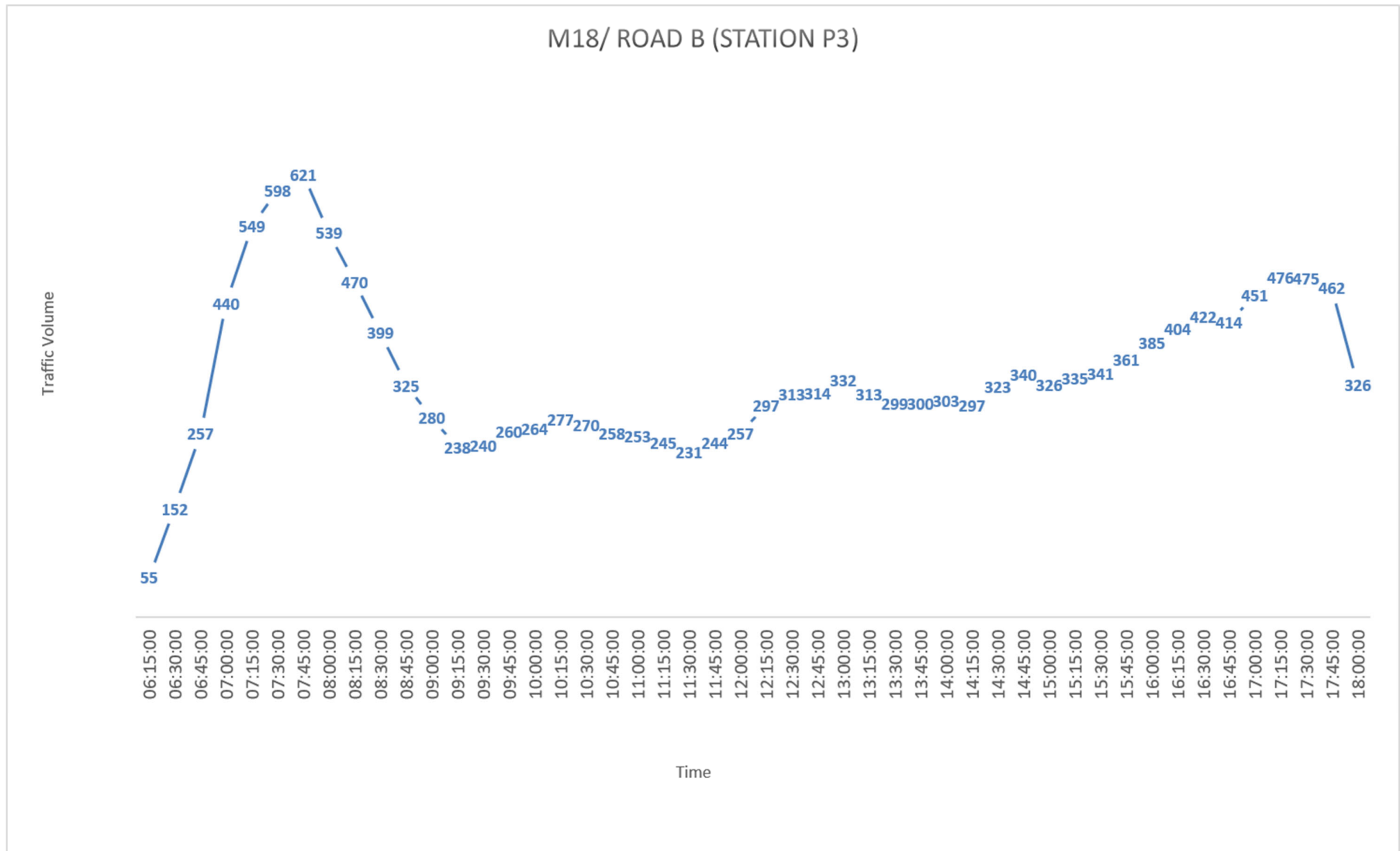


Figure 4-4: Hourly Traffic Volumes (Intersection 3)

It can be noted in **Figure 4-4** that for the M18/ Road B intersection (Intersection 3), the morning peak hour is from 07h00 to 08h00 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 621 and 476 vehicles per hour for the morning and afternoon peaks respectively.

Table 4-5: Peak hour traffic volumes at M18/ Road C (Intersection 4)

APPROACH	Morning Peak Volumes (07h00 to 08h00)			
	Left	Through	Right	TOTAL
Road C (Northbound)	3	-	3	6
M18 (Westbound)	10	145	-	155
M18 (Eastbound)	-	412	6	418
APPROACH	Afternoon Peak Volumes (16h30 to 17h30)			
	Left	Through	Right	TOTAL
Road C (Northbound)	0	-	6	6
M18 (Westbound)	20	220	-	240
M18 (Eastbound)	-	130	5	135

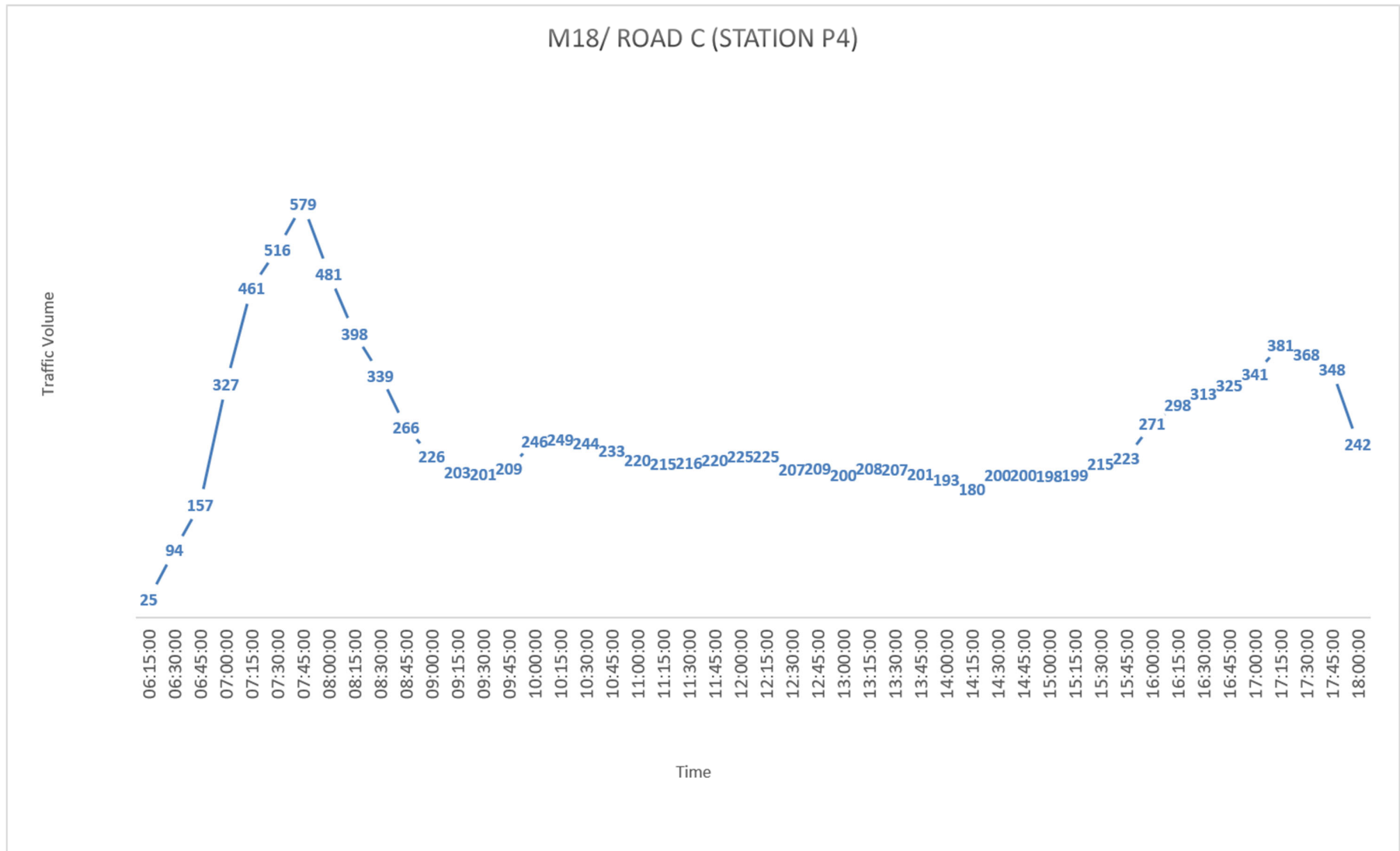


Figure 4-5: Hourly Traffic Volumes (Intersection 4)

It can be noted in **Figure 4-5** that for the M18/ Road C intersection (Intersection 4), the morning peak hour is from 07h00 to 08h00 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 579 and 381 vehicles per hour for the morning and afternoon peaks respectively.

Table 4-6: Peak hour traffic volumes at M18/ Glendore Road (Intersection 5)

APPROACH	Morning Peak Volumes (07h00 to 08h00)			
	Left	Through	Right	TOTAL
Glendore Road (Southbound)	219	-	131	350
M18 (Westbound)	-	79	70	149
M18 (Eastbound)	38	194	-	232
APPROACH	Afternoon Peak Volumes (16h30 to 17h30)			
	Left	Through	Right	TOTAL
Glendore Road (Southbound)	71	-	39	110
M18 (Westbound)	-	83	128	211
M18 (Eastbound)	80	65	-	145

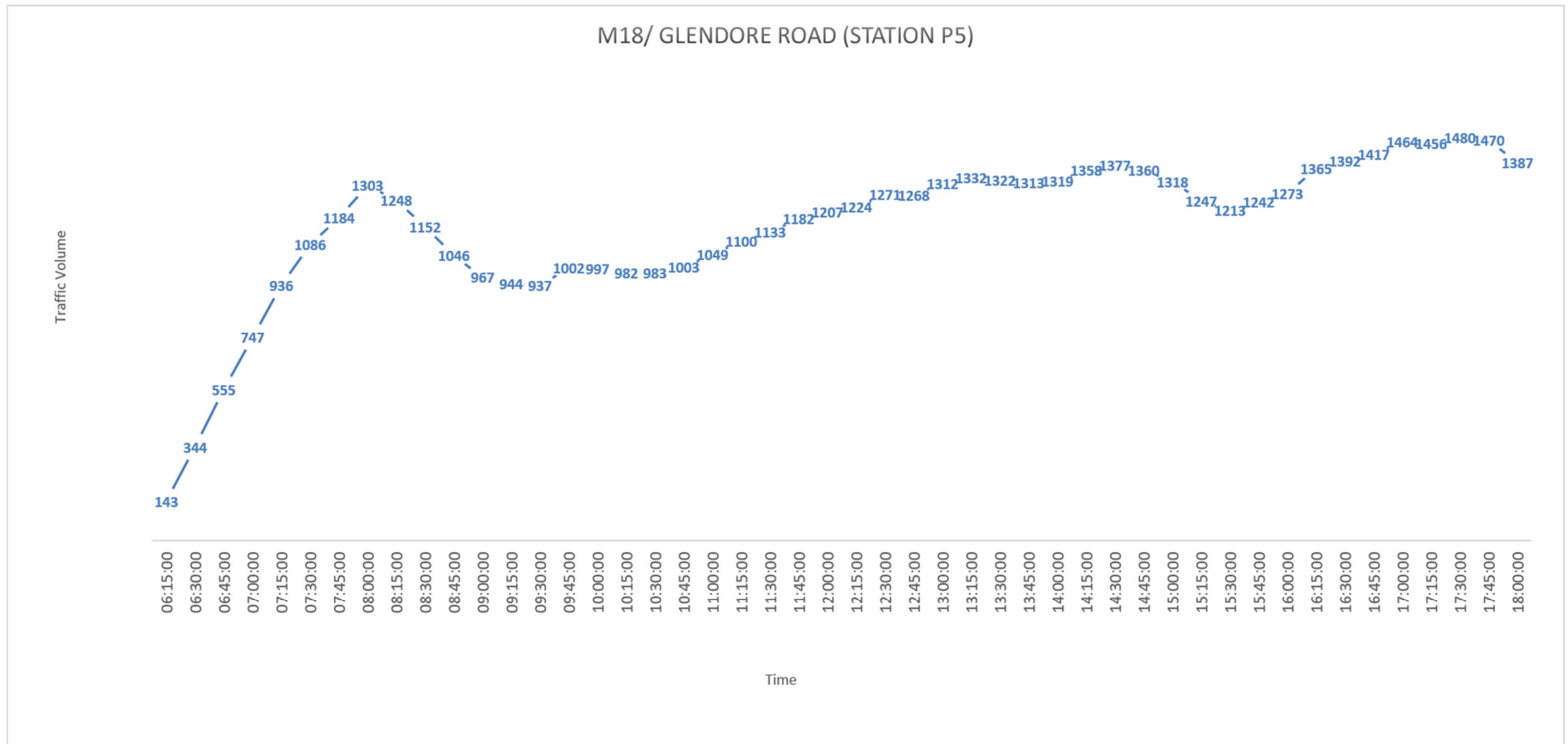


Figure 4-6: Hourly Traffic Volumes (Intersection 5)

It can be noted in **Figure 4-6** that for the M18/ Glendore Road intersection (Intersection 5), the morning peak hour is from 07h00 to 08h00 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 1303 and 1480 vehicles per hour for the morning and afternoon peaks respectively.

Table 4-7: Peak hour traffic volumes at M18/ Access to Smartstone Access (Intersection 6)

APPROACH		Morning Peak Volumes (07h00 to 08h00)			
		Left	Through	Right	TOTAL
Smartstone Access (Southbound)		0	-	1	1
M18 (Eastbound)		2	420	-	422
M18 (Westbound)		-	160	3	163
APPROACH		Afternoon Peak Volumes (16h30 to 17h30)			
		Left	Through	Right	TOTAL
Smartstone Access (Southbound)		0	-	0	0
M18 (Eastbound)		0	147	-	147
M18 (Westbound)		-	255	0	255

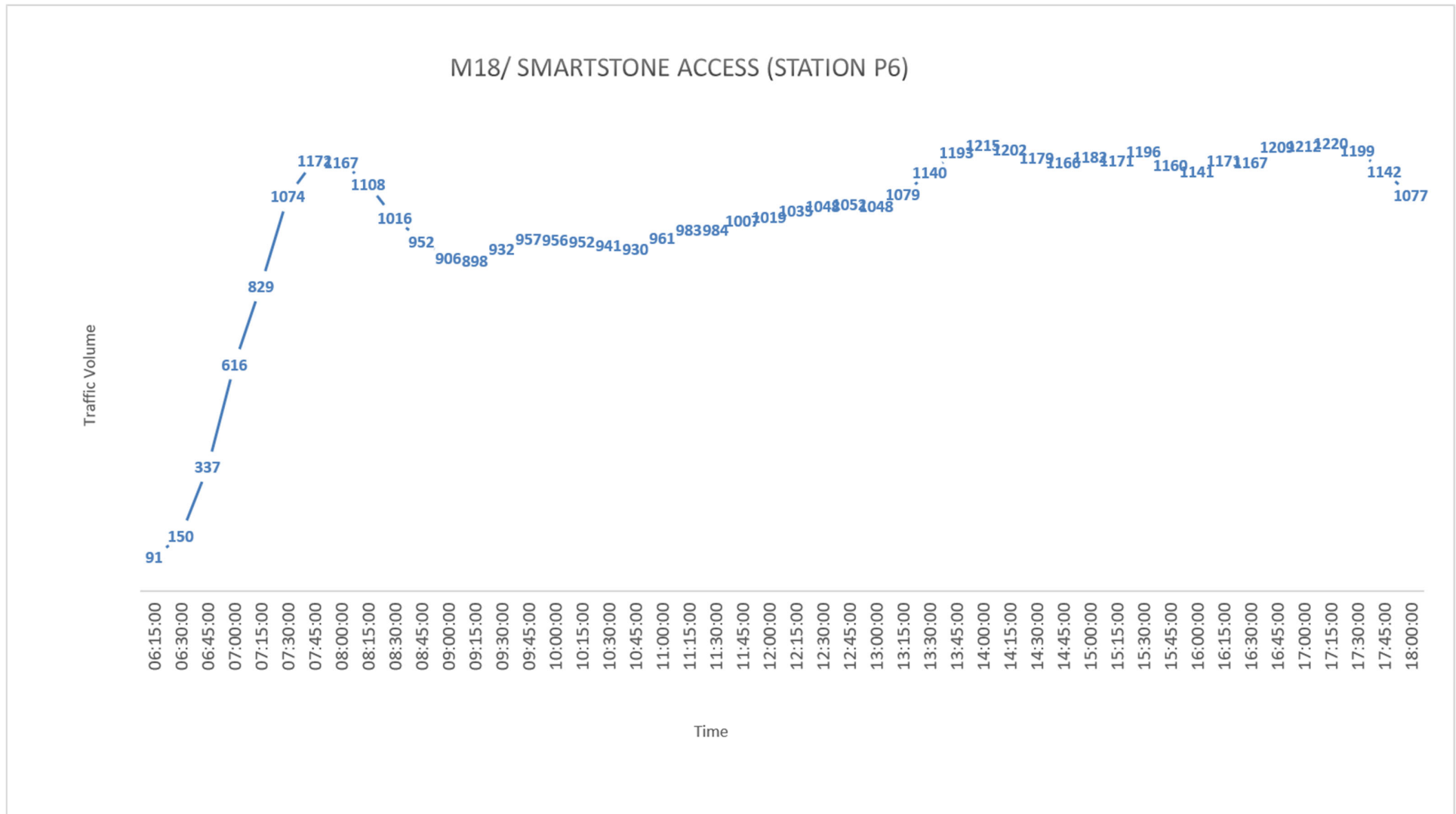


Figure 4-7: Hourly Traffic Volumes (Intersection 6)

It can be noted in **Figure 4-7** that for the M18/ Smartstone Access intersection (Intersection 6), the morning peak hour is from 07h00 to 08h00 and the afternoon peak is from 16h30 to 17h30. The intersection peak hour volumes are 1172 and 1220 vehicles per hour for the morning and afternoon peaks respectively.

5. TOWNSHIP ROAD NETWORK AND ACCESSIBILITY

Provision has been made for two vehicular access points to serve the proposed development. Access to the southern portion of the site will be provided directly from Victoria Drive, while the northern portion will be accessed via the existing informal road connecting to Beethoven Avenue.

No vehicular connection is proposed between the northern and southern portions of the development. However, a dedicated pedestrian link will be provided adjacent to the business development component of the site to facilitate safe and convenient pedestrian movement between the two areas. This connection is intended to accommodate pedestrian activity, particularly that of school learners travelling between the northern and southern sections of the development. The proposed access arrangements are illustrated in **Figure 5-1**.

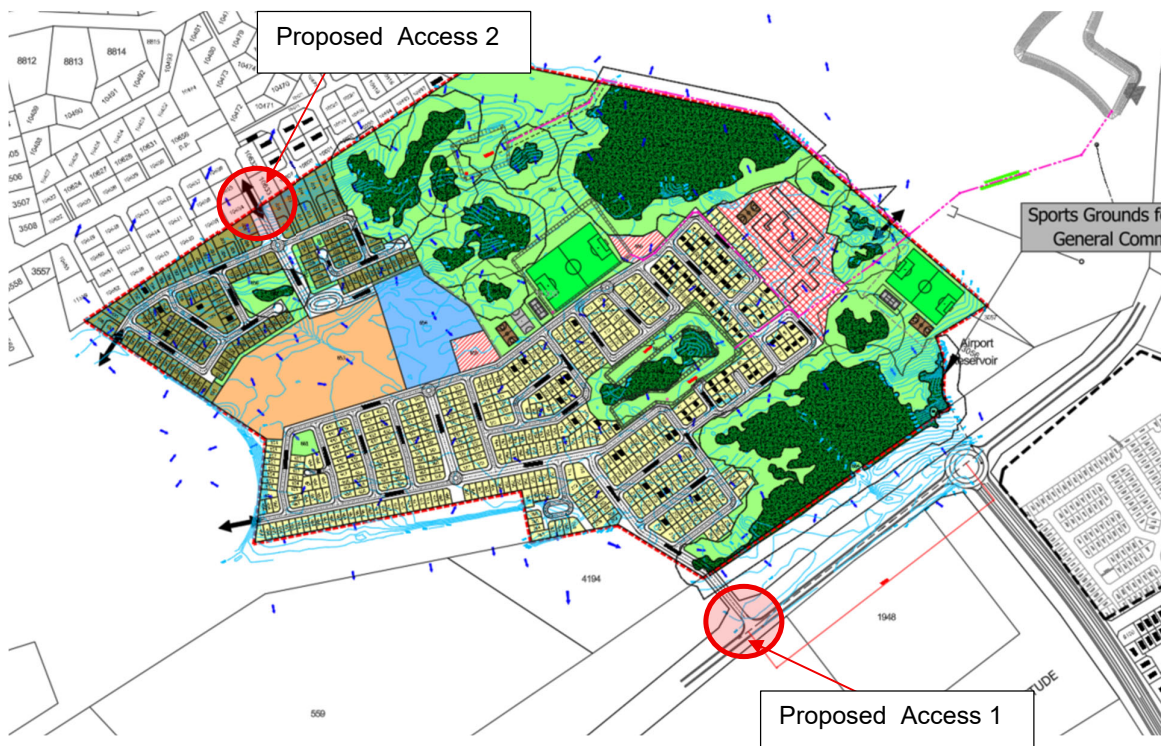


Figure 5-1: Development Access Points

In accordance with the recommendations of the TRH 26: South African Road Classification and Access Management Manual, Class 4 urban roads typically require road reserve widths ranging from 16m to 40m, while Class 5 urban roads generally require reserves of between 10m and 25m. The internal road network has therefore been planned to align with the principles and recommendations of TRH 26, to the extent that is practically achievable within the site constraints.

The primary access roads are proposed with a 6m wide surfaced carriageway within a 16m road reserve. The remaining internal roads will also be surfaced and will be accommodated within 1m road reserves, providing adequate space for the carriageway, municipal services, stormwater infrastructure, and pedestrian facilities where required.

6. TRAFFIC DEMAND

The traffic demand for the development will take into consideration two horizon years. The design and the planning horizon years. The design horizon year is the year selected for determining transportation improvements that are required to accommodate the proposed development with the prevailing traffic conditions. The transportation/ road improvements must be designed for a horizon year of 5 years. The planning horizon year, though not used for determining the transportation improvements that are required for the development, is the year selected for determining whether it is physically possible to accommodate the proposed development together with future traffic growth. The planning horizon year to be adopted for the development is 10 years (Committee of Transport Officials (COTO), 2012).

6.1 Trip Generation

Business I and institutional I land uses are expected to be subservient to the residential landuse and assumed only to generate internal trips within the development. The impact of the proposed development on the external road network is 847 (AM Peak), 847 (PM Peak) and 432 (SAT Peak) peak hour trips. The weekday AM, PM and Saturday peak hour trip generation for the proposed development is shown in **Table 6-1**, **Table 6-2** and **Table 6-3** respectively.

Table 6-1: Weekday AM Trip Generation

MORNING PEAK HOUR								
						TRIPS		
Township	Land-Use	Extent (Units)	Trip Rate	Split		Trip Generations		
				In	Out	In	Out	Total
Erf 11305 Walmer	Residential I (FBO)	509	1.00	25%	75%	127	382	509
	Residential I (GH)	143	1.00	25%	75%	36	107	143
	Residential IV	300	0.65	25%	75%	49	146	195
TRIPS GENERATED						212	635	847

Table 6-2: Weekday PM Trip Generation

AFTERNOON PEAK HOUR								
						TRIPS		
Township	Land-Use	Extent (Units)	Trip Rate	Split		Trip Generations		
				In	Out	In	Out	Total
Erf 11305 Walmer	Residential I (FBO)	509	1.00	70%	30%	356	153	509
	Residential I (GH)	143	1.00	70%	30%	100	43	143
	Residential IV	300	0.65	70%	30%	137	58	195
TRIPS GENERATED						593	254	847

Table 6-3: Weekend Trip Generation

SATURDAY PEAK HOUR								
						TRIPS		
Township	Land-Use	Extent (Units)	Trip Rate	Split		Trip Generations		
				In	Out	In	Out	Total
Erf 11305 Walmer	Residential I (FBO)	509	0.50	50%	50%	127	127	254
	Residential I (GH)	143	0.50	50%	50%	36	36	72
	Residential IV	300	0.35	50%	50%	53	53	106
TRIPS GENERATED						216	216	432

6.2 Trip Distribution and Assignments

6.2.1 Background

The distribution of development-generated traffic was determined based on the assumption that future travel patterns will broadly reflect the prevailing traffic movements within the surrounding road network.

Accordingly, the generated trips have been distributed in proportion to the observed directional traffic flows on the adjacent road network. The assignment also takes into consideration the location of the proposed access points, the hierarchy and function of the surrounding roads, expected travel desire lines, and the most convenient routes available to motorists travelling to and from the development.

6.2.2 Trip Distribution and Assignment

Traffic generated by the developments located within the southern portion of the site has been assigned to the proposed southern access (Access 1) from Victoria Drive, while traffic associated with the northern portion of the development has been assigned to the proposed northern access (Access 2), which connects to the existing informal road linking to Beethoven Avenue. Trip distribution and assignment were undertaken in accordance with the anticipated origins and destinations of development-generated traffic, taking into account the most logical and efficient travel routes available within the surrounding road network.

6.3 Traffic Growth

According to the Nelson Mandela Bay Metropolitan Municipality Integrated Development Plan (IDP) 2024/25, the local economy is primarily driven by the manufacturing, finance, community services, trade and transport sectors, with the automotive industry continuing to play a significant role in regional economic activity. Although the municipality has experienced relatively slow economic growth over the past decade, economic forecasts project an average annual growth rate of approximately 1.5% between 2021 and 2026, supported by continued investment in the manufacturing, logistics and automotive sectors.

In view of the municipality's moderate population and economic growth trends, together with the established nature of the surrounding road network, a conservative annual background traffic growth rate of 2% has been adopted for the assessment. This growth rate is considered appropriate for forecasting future traffic conditions over the assessment horizon and provides a reasonable basis for evaluating the operational performance of the surrounding road network under future-year scenarios.

The estimated 2026 morning and afternoon peak-hour traffic volumes, together with the projected 2030 and 2035 peak-hour traffic volumes, are presented in **Annexure B**.

7. TRAFFIC IMPACT AND CAPACITY ANALYSES

7.1 Introduction

Capacity analysis for the proposed development was done using SIDRA intersection software to determine the impact and intersection upgrades required. Traffic volumes recorded during the AM and PM peak periods were used as input parameters to assess the operational performance of the affected intersections.

7.2 Analysis Scenarios

Traffic impact assessment was undertaken for the following scenarios:

- Status quo
- Design horizon year (5 year assessment)
- Planning horizon year (10 year assessment)

The status quo assessment was undertaken to understand what the current Levels of Service are at the intersections near the proposed development. The design horizon year assessment was undertaken with the purpose of establishing mitigation measures that are required to accommodate the development where required.

The following four scenarios were analysed as part of this assessment:

- Scenario 1: Status quo (2026 Background Traffic Demand)
- Scenario 2: 2031 Background Traffic Demand
- Scenario 3: 2031 Background + Development Traffic Demand
- Scenario 4: 2036 Background + Development Traffic Demand

The table below presents the results of the capacity analysis for the critical intersections and proposed access points intended to serve the development. Detailed capacity analysis outcomes for Scenarios 1, 2, 3, and 4 are provided in **Annexures C, D, E, and F** respectively.

Table 7-1: Capacity Analysis Results

No	Intersection	Performance	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
			Weekday Morning Peak Hour	Weekday Afternoon Peak Hour	Weekday Morning Peak Hour	Weekday Afternoon Peak Hour	Weekday Morning Peak Hour	Weekday Afternoon Peak Hour	Weekday Morning Peak Hour	Weekday Afternoon Peak Hour
1	Beethoven Avenue/ Road A	V/C Ratio	0.011	0.008	0.012	0.008	0.085	0.062	0.086	0.063
		Delays	1.5	2.3	1.4	2.1	6.4	5.5	6.3	5.5
		LOS	A	A	A	A	A	A	A	A
2	Schubert Road/ Beethoven Avenue	V/C Ratio	0.026	0.066	0.029	0.073	0.071	0.075	0.073	0.082
		Delays	5.8	6.6	5.8	6.7	5.7	6.5	5.7	6.5
		LOS	A	A	A	A	A	A	A	A
3	M18/ Road B	V/C Ratio	0.229	0.158	0.253	0.174	0.253	0.174	0.279	0.193
		Delays	1.0	1.3	1.1	1.4	1.1	1.4	1.1	1.4
		LOS	A	A	A	A	A	A	A	A
4	M18/ Road C	V/C Ratio	0.227	0.130	0.251	0.144	0.251	0.144	0.276	0.158
		Delays	0.3	0.6	0.3	0.6	0.3	0.6	0.3	0.6
		LOS	A	A	A	A	A	A	A	A
5	M18/ Glendore Road	V/C Ratio	0.370	0.132	0.423	0.146	0.443	0.168	0.508	0.190
		Delays	5.6	4.7	5.9	4.7	6.2	4.8	6.7	4.9
		LOS	A	A	A	A	A	A	A	A
6	M18/ Smartstone Access	V/C Ratio	0.228	0.138	0.252	0.153	0.858	0.431	0.940	0.452
		Delays	0.1	0.1	0.1	0.1	10.2	5.4	13.1	5.5
		LOS	A	A	A	A	B	A	B	A

The assessment indicates that the intersections will continue to operate within acceptable capacity following the introduction of traffic generated by the proposed development. It should be noted that Intersection 3 (M18/ Road B) is planned to be upgraded to a traffic roundabout as part of a separate infrastructure initiative. This upgrade is independent of the proposed Erf 11305 Walmer development and has therefore not been identified as a mitigation measure required to accommodate the Erf 11305 Walmer development-generated traffic.

8. PROPOSED UPGRADES

Intersection upgrades required to accommodate the background traffic, and the future traffic volumes on the existing road network are shown and described in **Table 8-1** below.

Table 8-1: Proposed Upgrades

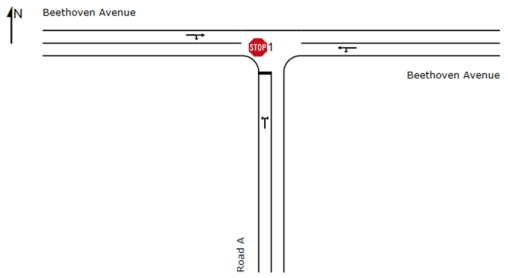
Nr	Intersection Name	Scenario 1: 2026 Background Traffic Volumes	Scenario 2: 2031 Background Traffic Volumes	Scenario 3: 2031 Background + Development Traffic Volumes	Scenario 4: 2036 Background + Development Traffic Volumes
1.	Beethoven Avenue/ Road A		The existing road layout has sufficient capacity to accommodate the anticipated background traffic growth. It is, however, recommended that the southern leg be paved to improve operational efficiency, road safety, and overall network continuity.	The existing road layout has sufficient capacity to accommodate the anticipated background traffic growth as well as the development traffic.	The existing layout can accommodate the traffic demand.

Table 8-1: Proposed Upgrades

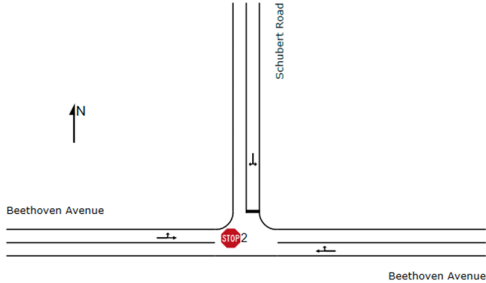
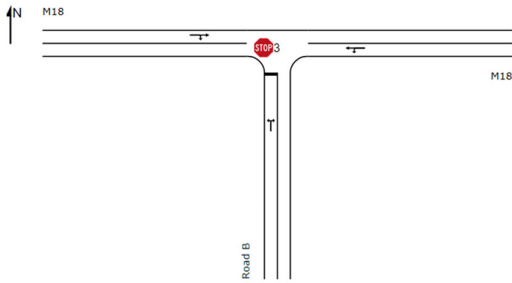
Nr	Intersection Name	Scenario 1: 2026 Background Traffic Volumes	Scenario 2: 2031 Background Traffic Volumes	Scenario 3: 2031 Background + Development Traffic Volumes	Scenario 4: 2036 Background + Development Traffic Volumes
2.	Schubert Road/ Beethoven Avenue		The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.
3.	M18/ Road B		The existing intersection configuration has sufficient capacity to accommodate the anticipated background traffic growth. However, the intersection is also expected to serve as an access point to the future development planned within the Walmer Area G-West 2 precinct. This intersection has been upgraded to a roundabout. The proposed roundabout will provide improved operational efficiency, increased capacity, and improved traffic flow while	The proposed roundabout layout has sufficient capacity to accommodate the traffic demand.	The proposed roundabout has sufficient capacity to accommodate the traffic demand.

Table 8-1: Proposed Upgrades

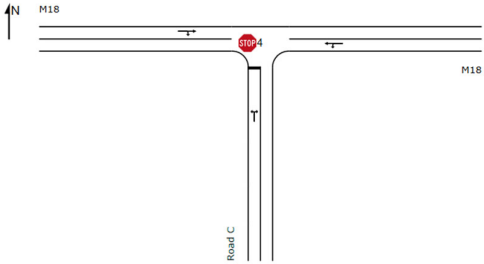
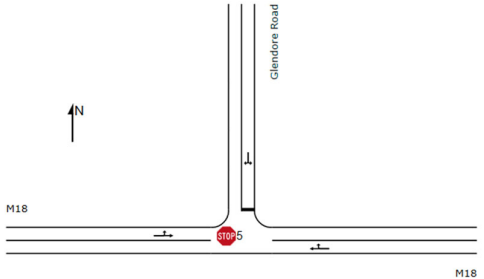
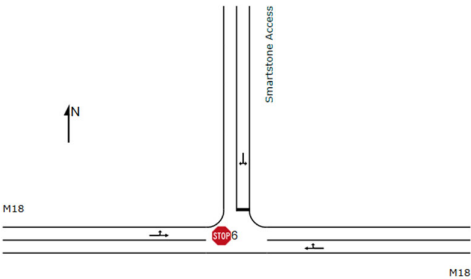
Nr	Intersection Name	Scenario 1: 2026 Background Traffic Volumes	Scenario 2: 2031 Background Traffic Volumes	Scenario 3: 2031 Background + Development Traffic Volumes	Scenario 4: 2036 Background + Development Traffic Volumes
			safely accommodating future traffic volumes.		
4.	M18/ Road C		The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.
5.	M18/ Glendore Road		The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.

Table 8-1: Proposed Upgrades

Nr	Interse ction Name	Scenario 1: 2026 Background Traffic Volumes	Scenario 2: 2031 Background Traffic Volumes	Scenario 3: 2031 Background + Development Traffic Volumes	Scenario 4: 2036 Background + Development Traffic Volumes
6.	M18/ Smartstone Access/ Proposed Access		This access will serve the proposed southern portion of the development. The existing intersection is expected to operate within acceptable capacity under anticipated background traffic growth conditions. However, to adequately accommodate development-generated traffic and improve the overall standard of access, it is recommended that the northern approach be upgraded to a surfaced roadway.	The existing road layout has sufficient capacity to accommodate the traffic demand.	The existing road layout has sufficient capacity to accommodate the traffic demand.

9. PUBLIC TRANSPORT AND NON-MOTORISED TRANSPORT

An assessment of public transport (PT) and non-motorised transport (NMT) conditions was undertaken within the study area to better understand existing travel patterns, identify infrastructure gaps, and inform recommendations that align with National and Provincial PT and NMT planning guidelines. The assessment included public transport and pedestrian count surveys, field observations of transport operations, and a review of surrounding land uses to understand how people currently move through the area and where improvements are most needed.

Field observations showed that the study area has very limited formal PT and NMT infrastructure. The traffic counts showed that the study area is characterised high proportion of light vehicles and little to no public transport. The figure below shows the proportion of public transport compared to other modes of transport at the surveyed intersections.

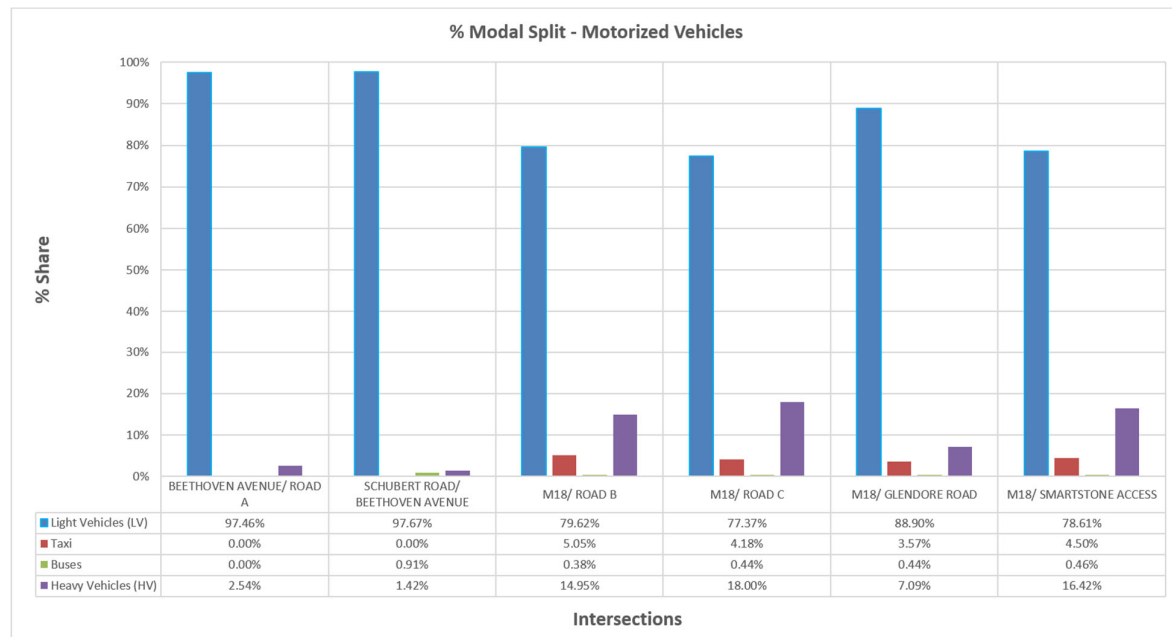


Figure 9-1: Modal Split

Pedestrian activity was also observed in the study area, despite the lack of formal walking facilities. Pedestrians currently walk along gravel shoulders, road reserves, and informal footpaths because continuous paved sidewalks are largely absent. On the higher-order roads (M18), where vehicle speeds are generally higher, pedestrians are often forced to walk close to fast-moving traffic, increasing their exposure to potential conflicts with vehicles. The figure below shows the pedestrian volumes at the surveyed intersections.

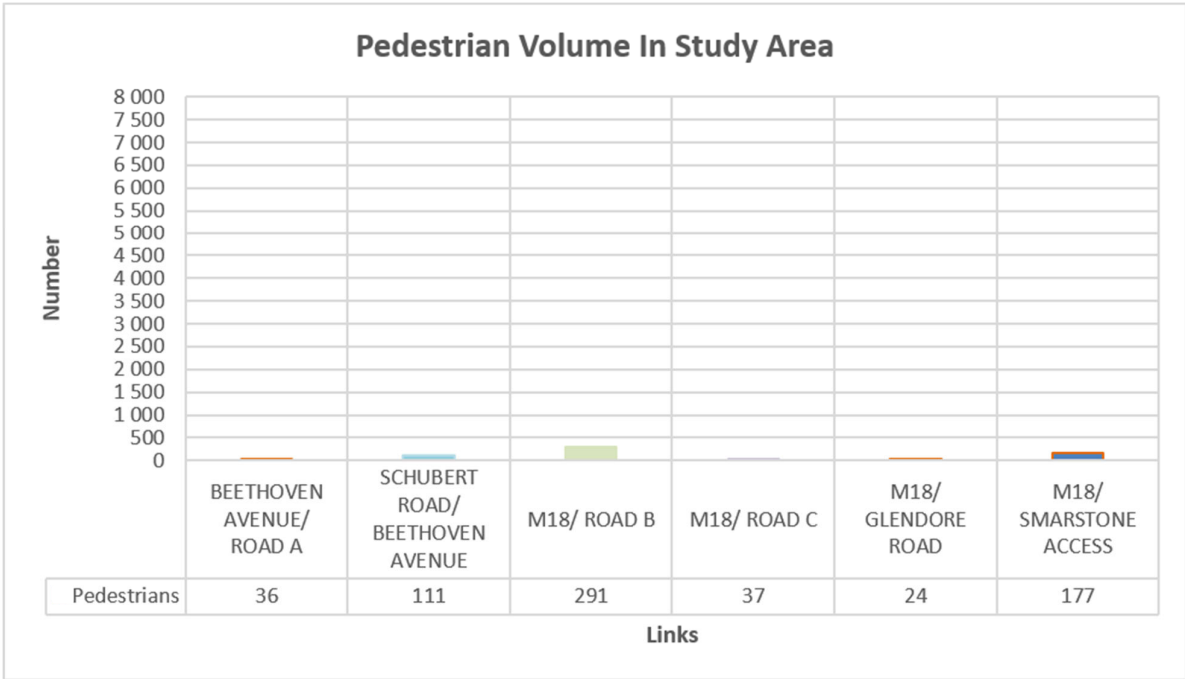


Figure 9-2: Pedestrian Volumes

The assessment further highlighted the limited provision of supporting infrastructure such as pedestrian crossings, public transport shelters, paved waiting areas, and facilities that accommodate people with disabilities. The absence of these features affects overall accessibility and particularly impacts vulnerable users such as children, elderly residents, and persons with reduced mobility.

9.1 Proposed Public Transport Facilities

According to TMH 16: *Traffic Impact Assessment Study Manual, Volume 2*, public transport facilities should be located within a reasonable walking distance of major trip generators, attractors, and modal transfer points to ensure convenient and accessible access to public transport services.

Based on the anticipated public transport and pedestrian demand associated with the proposed development, the following PT facilities are recommended to improve accessibility, enhance user safety, and support the efficient operation of public transport services within the study area:

- Sheltered public transport lay-bys are proposed at the primary access point along the proposed M18 roadway. These facilities will provide safe and convenient areas for public transport vehicles to stop without impeding through traffic, while also offering passengers a dedicated waiting area protected from adverse weather conditions.
- To further improve the functionality of the proposed public transport facilities, it is recommended that the lay-bys be complemented by supporting infrastructure such as passenger shelters, paved waiting areas, pedestrian walkways, appropriate signage, and lighting. The provision of these facilities will improve user comfort, accessibility, and safety, particularly for vulnerable users such as school children, elderly persons, and individuals with mobility impairments.
- The proposed public transport facility should also be integrated with the planned non-motorised transport network to ensure safe and direct pedestrian connections between residential areas, public transport stops, and other key destinations within the development

9.2 Proposed Non-Motorised Transport Facilities

To support existing and future pedestrian movement within the study area, and to promote safe, accessible, and sustainable travel, the following non-motorised transport (NMT) facilities are recommended. These recommendations are informed by observed pedestrian activity, anticipated demand associated with the proposed development, and applicable NMT planning and design guidelines:

- **Sidewalks:** Continuous pedestrian sidewalks should be provided throughout the development to create a safe and accessible walking environment. In accordance with the NMT Facility Guidelines, sidewalks should have a minimum unobstructed width of 1.5m, although wider facilities should be considered in areas where higher pedestrian volumes are anticipated (i.e the M18 route).

At a minimum, sidewalks should be provided along both sides of the collector roads serving the development, as well as at all site access points to ensure safe connections between the development and the surrounding road network. Sidewalks should be designed to provide continuous links between residential areas, public transport facilities, schools, community facilities, commercial areas, and other key destinations.

- **Universal Access (UA) Design:** Universal Access (UA) principles should be incorporated into the design of all pedestrian facilities to ensure that the transport network is accessible to all users, including persons with disabilities, the elderly, children, and individuals with reduced mobility. At intersections and pedestrian crossing points, universal access measures should include dropped kerbs, tactile paving, appropriate ramp gradients, and adequate crossing widths. The incorporation of UA principles will improve accessibility,

promote social inclusion, and ensure compliance with current accessibility standards and best-practice design guidelines.

- Traffic calming measures: Given the observed and anticipated pedestrian activity within the development, particularly along routes connecting residential areas to schools, public transport facilities, and other community amenities, traffic calming measures should be implemented at these roads to improve pedestrian safety and reduce vehicle operating speeds.

It is recommended that traffic calming interventions, including speed humps, speed tables, raised pedestrian crossings, and other suitable measures, be considered on residential streets that are expected to function as key pedestrian desire lines. The selection and design of these measures should be undertaken in accordance with the National Guidelines for Traffic Calming and should be appropriate to the functional classification and operating characteristics of the road concerned/

10. CONCLUSIONS AND RECOMMENDATIONS

10.1 Conclusions

Based on the findings presented in this report, the following conclusions have been drawn:

1. The Nelson Mandela Bay Municipality appointed Gwaca Consulting cc to conduct a Traffic Impact Assessment (TIA) for the proposed rezoning and subdivision of Erf 11305 Walmer covering an area of 43.73 hectares, located within the jurisdiction of the Nelson Mandela Municipality in the Eastern Cape Province.
2. Traffic counts were conducted by Gwaca Consulting on the 02nd of June 2026.
3. The proposed development is projected to generate approximately 847 vehicle trips during both the weekday AM and PM peak periods, and 432 vehicle trips during the Saturday peak period.
4. A traffic growth rate of 2% per annum was applied in this assessment.
5. The analysis was undertaken using SIDRA Intersection version 10.1 software to evaluate the operational performance of intersections and identify necessary upgrades to accommodate the proposed Hoedspruit township development, with a design horizon year of 2031 and a planning horizon year of 2036.
6. The operational assessment, inclusive of development-generated traffic, indicates that all intersections are expected to function at acceptable Levels of Service (LOS) under future conditions, provided that the recommended upgrades outlined in Section 8 are implemented.
7. Measures pertaining to public transport and non-motorised transport, as outlined in Sections 9 of this report, should be implemented to promote integrated and inclusive mobility solutions.

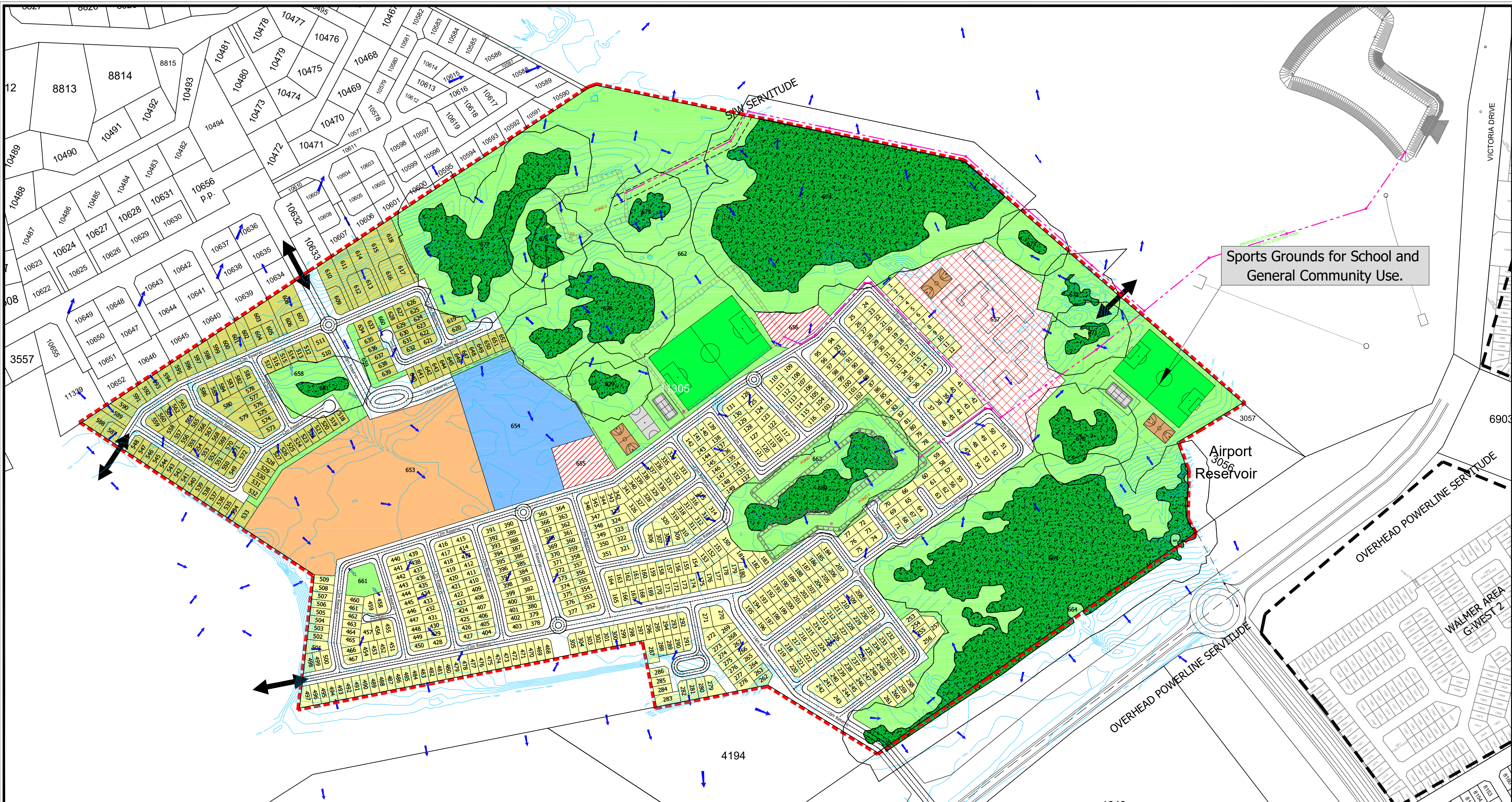
10.2 Recommendations

1. The proposed development should be considered favourably from a traffic engineering point of view by the relevant authorities.
2. Detailed designs for the proposed improvements should be undertaken by a professional engineer / technologist with suitable road design experience.

11. REFERENCES

1. Committee of Transport Officials (COTO). (2012). TMH 16 Volume 1 : South African Traffic Impact and Site Traffic Assessment Manual, Version 1.0. Pretoria: The South African National Roads Agency Limited.
2. COTO, TMH 17, South African Trip Data Manual, Version 1.01, September 2013.
3. Committee of Transport Officials (COTO). (2013). South African Road Classification and Access Management Manual (TRH 26). Pretoria: SANRAL.
4. National Department of Transport. (2003). PEDESTRIAN AND BICYCLE FACILITY GUIDELINES. Pretoria: National Department of Transport.
5. Nelson Mandela Bay Metropolitan Municipality Integrated Development Plan (IDP) 2024/25
6. Transportation Research Board, (2010) Highway Capacity Manual 2010, Washington, D.C, USA.

ANNEXURE A: TOWNSHIP LAYOUT



Sports Grounds for School and General Community Use.

Airport Reservoir

OVERHEAD POWERLINE SERVITUDE

WALMER AREA G-WEST 2

legend / key notes

- Subject Site
- Residential I (Free Basic Housing)
- Residential I (Gap Housing)
- Residential IV (± 124 units/ha)
- Business Zone I
- Institutional Zone I
- Authority Zone
- Open Space Zone I
- Special Use(Conservation Usage)
- Protected Forestry with 20m Buffer
- Pedestrian Walkway
- Stormwater Ponds

Zoning	No. Erven	Area in m ²	% Allo
Residential Zone IV	1	24236.23 m ²	5.54%
Residential Zone I (FBH)	509	81629.13 m ²	18.66%
Residential Zone I (Gap)	143	29478.47 m ²	6.74%
Business Zone I	1	11763.94 m ²	2.69%
Institutional Zone I	1	15598.56 m ²	3.57%
Authority Zone	2	3659.82 m ²	0.84%
Open Space Zone I	11	128761.40 m ²	29.44%
Special Use (Conservation Usage)	13	76015.22 m ²	17.38%
Transport Zone II	-	66130.23 m ²	15.15%
Total	681	437273.00 m ²	100.00%

Est. Residential Opportunities		No. Units	Area in m ²
Single Dwellings		652	111107.60 m ²
High Density Residential		300	24236.23 m ²
Total		952	135343.73 m ²
Gross Area	:	437273.00 m ² (43.73ha)	
Gross Residential Area (Single Dwellings)	:	111107.60 m ² (11.11ha)	
Gross Residential Density (Single Dwellings)	:	14.90 units / ha	
Nett Residential Density (Single Dwellings)	:	58.69 units / ha	
Average Erf Size (Single Dwellings)	:	170.41 m ²	

PROJECT

Proposed Rezoning & Subdivision of Erf 11305 Walmer

(Nelson Mandela Bay Municipality)

PREPARED BY:

METROPLAN

TOWN AND REGIONAL PLANNERS

P.O. Box 12313, Central, 6006 - 7 Upper Dickens Street, Central, Port Elizabeth, 6001 - Tel: +27 (0)41 373-1843 Fax: +27 (0)41 373-1838 E-mail: plan@metroplan.co.za

PROJECT LEADER

for metroplan: n. ndzombane

28/03/19

Date

DRAWN BY

L. Leeson

28/03/19

DESIGNED BY

n. ndzombane

28/03/19

DATE

SCALE

1 : 2 500 on A2

CLIENT

nelson mandela bay

PORT ELIZABETH | UITENHAGE | DESPATCH

CLIENT

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Proposed Rezoning and Subdivision Plan

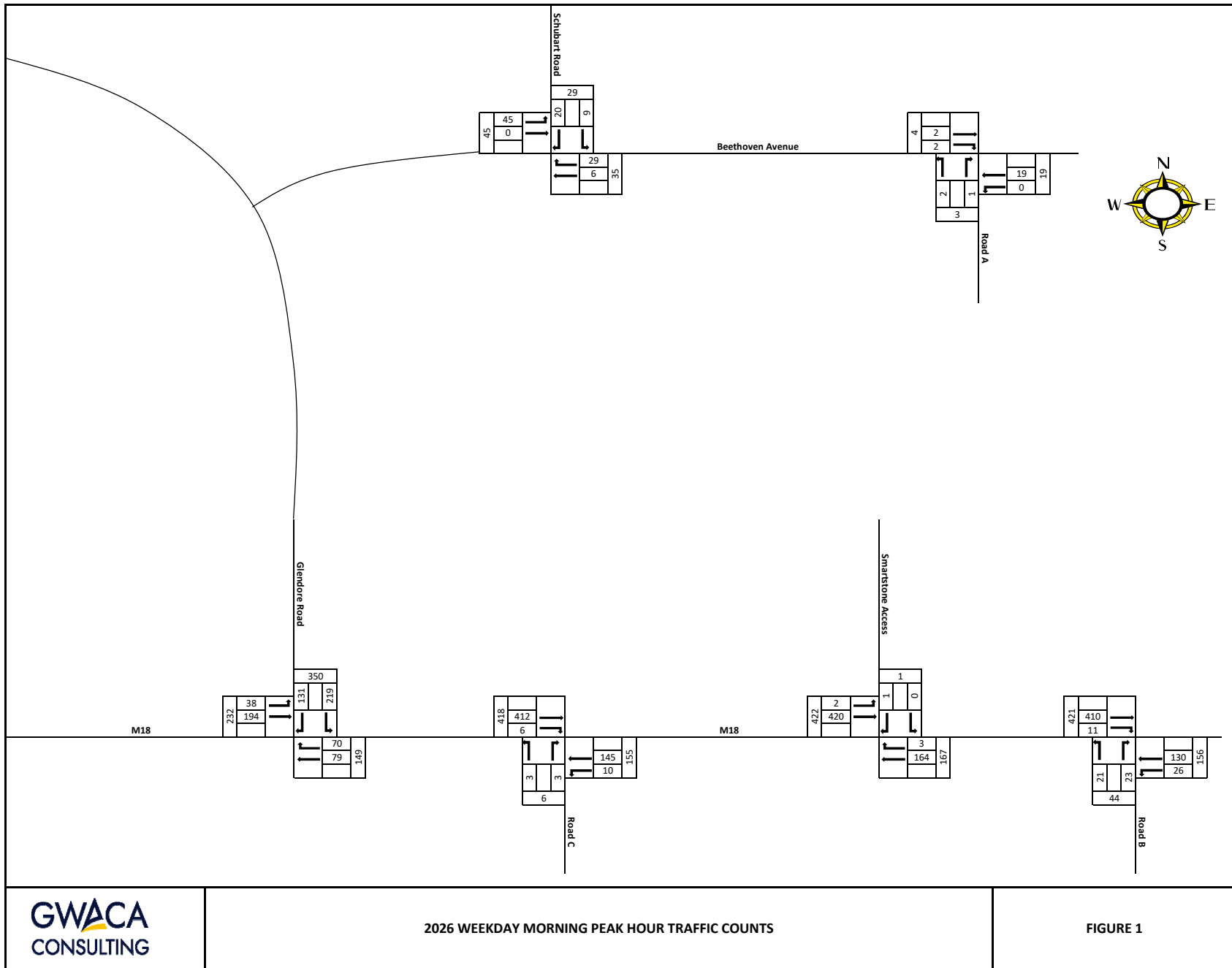
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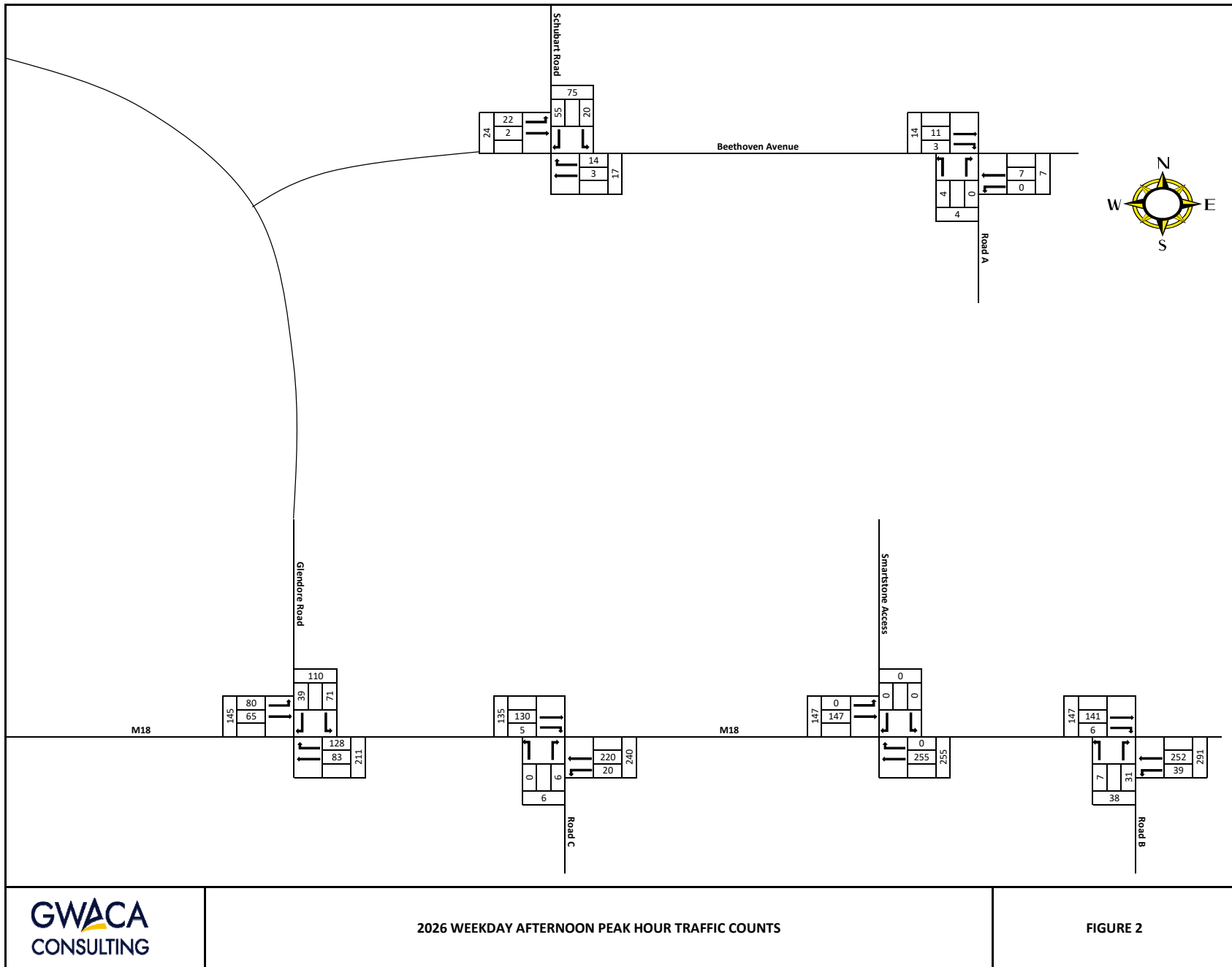
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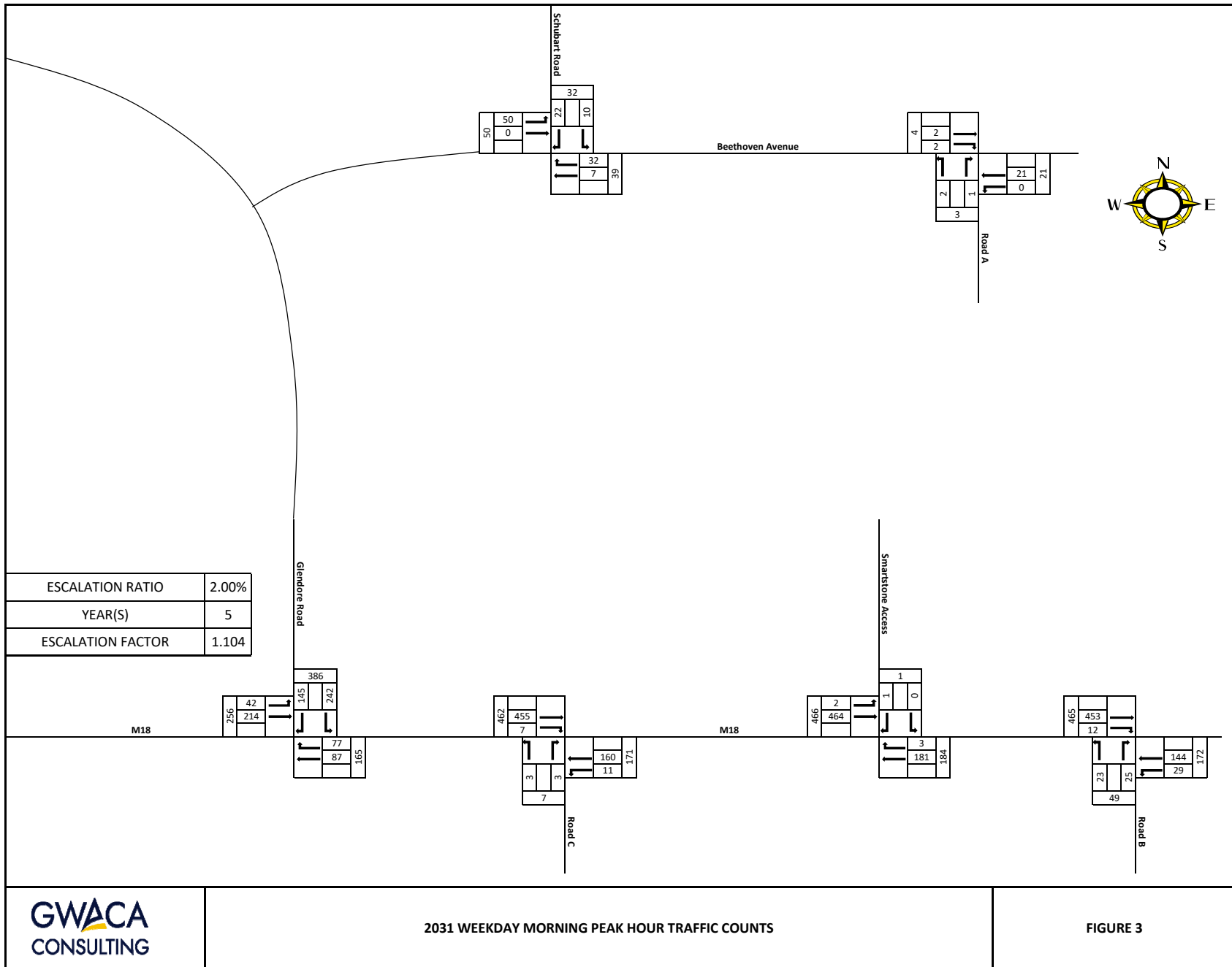
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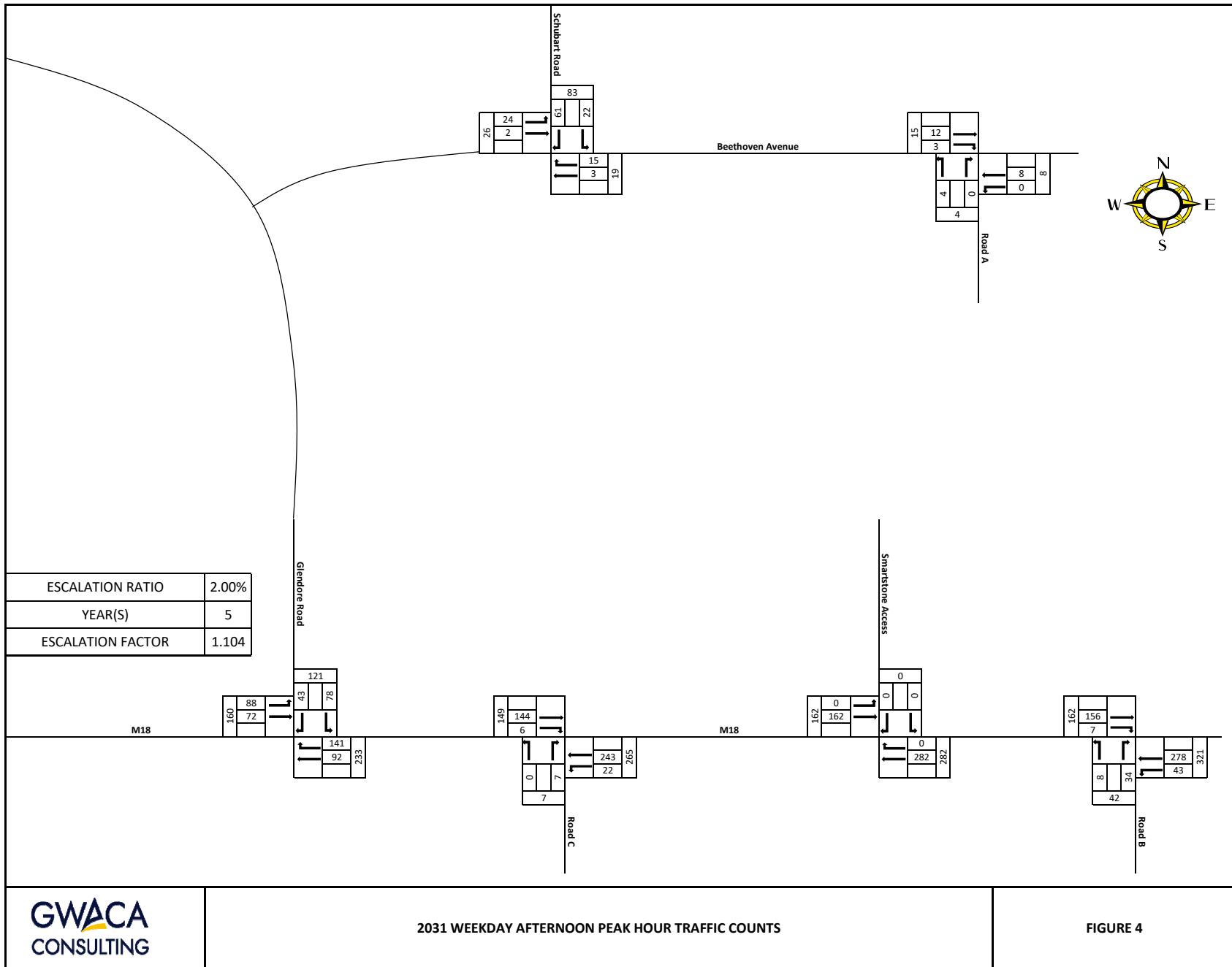
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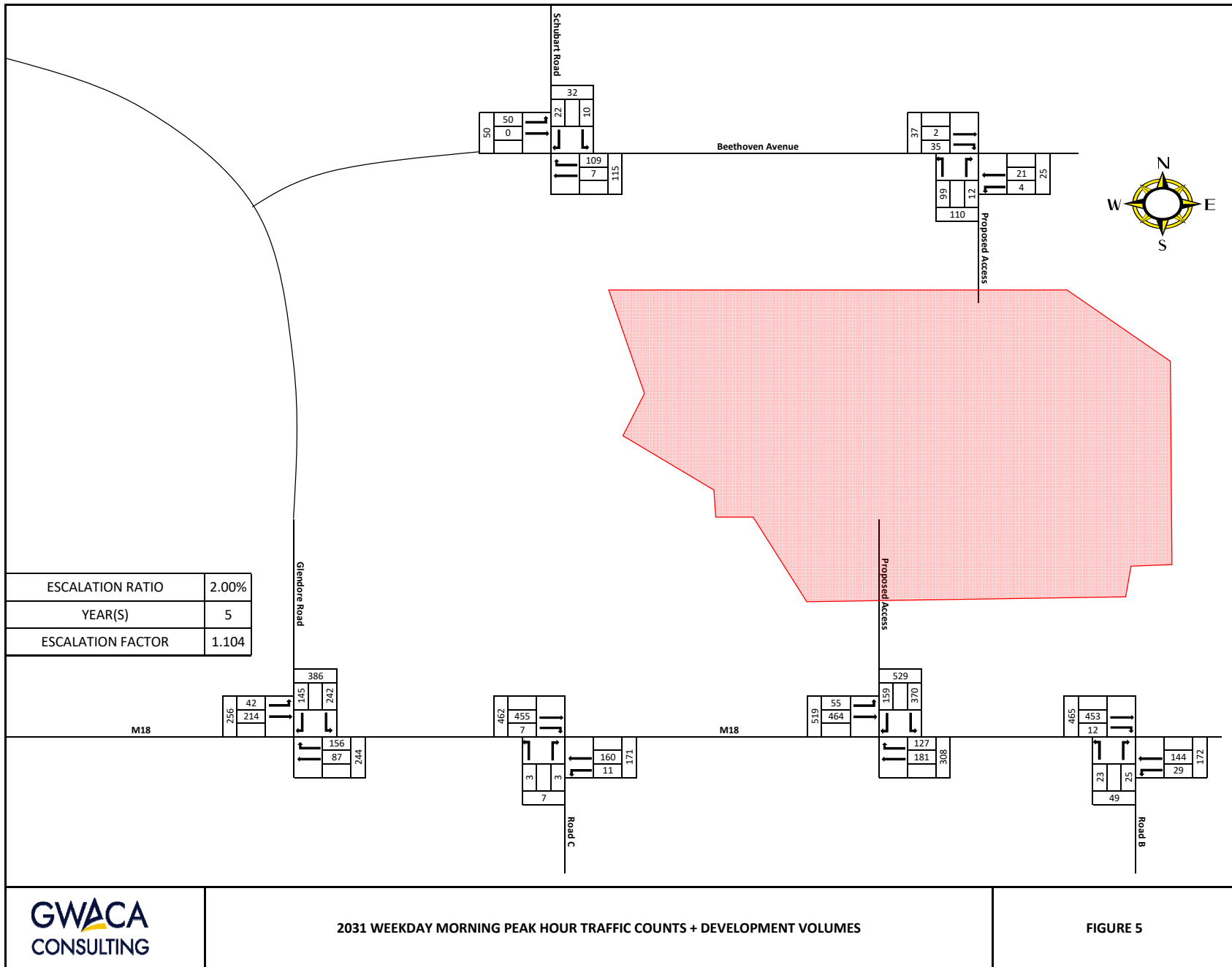
ANNEXURE B: TRAFFIC VOLUMES

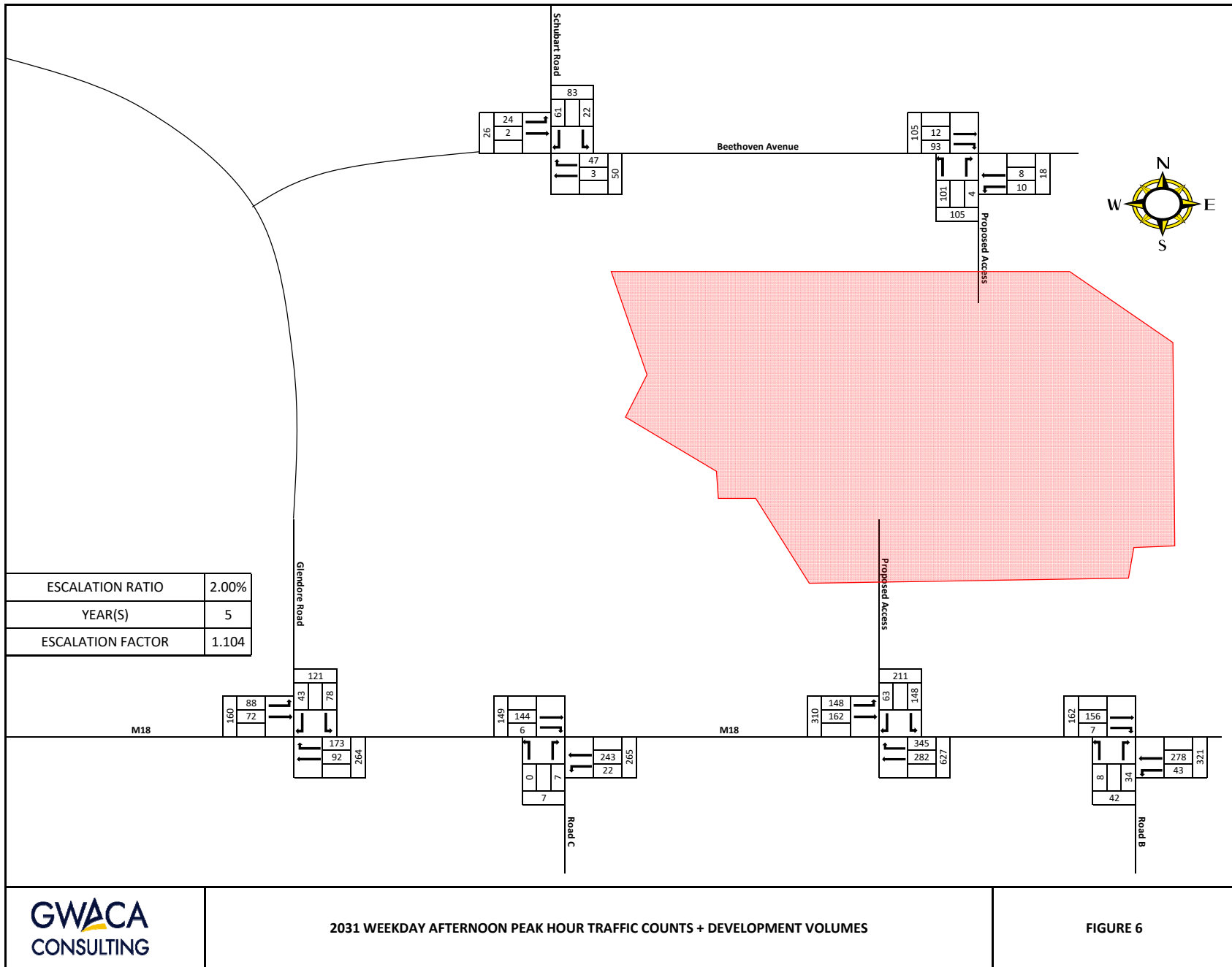


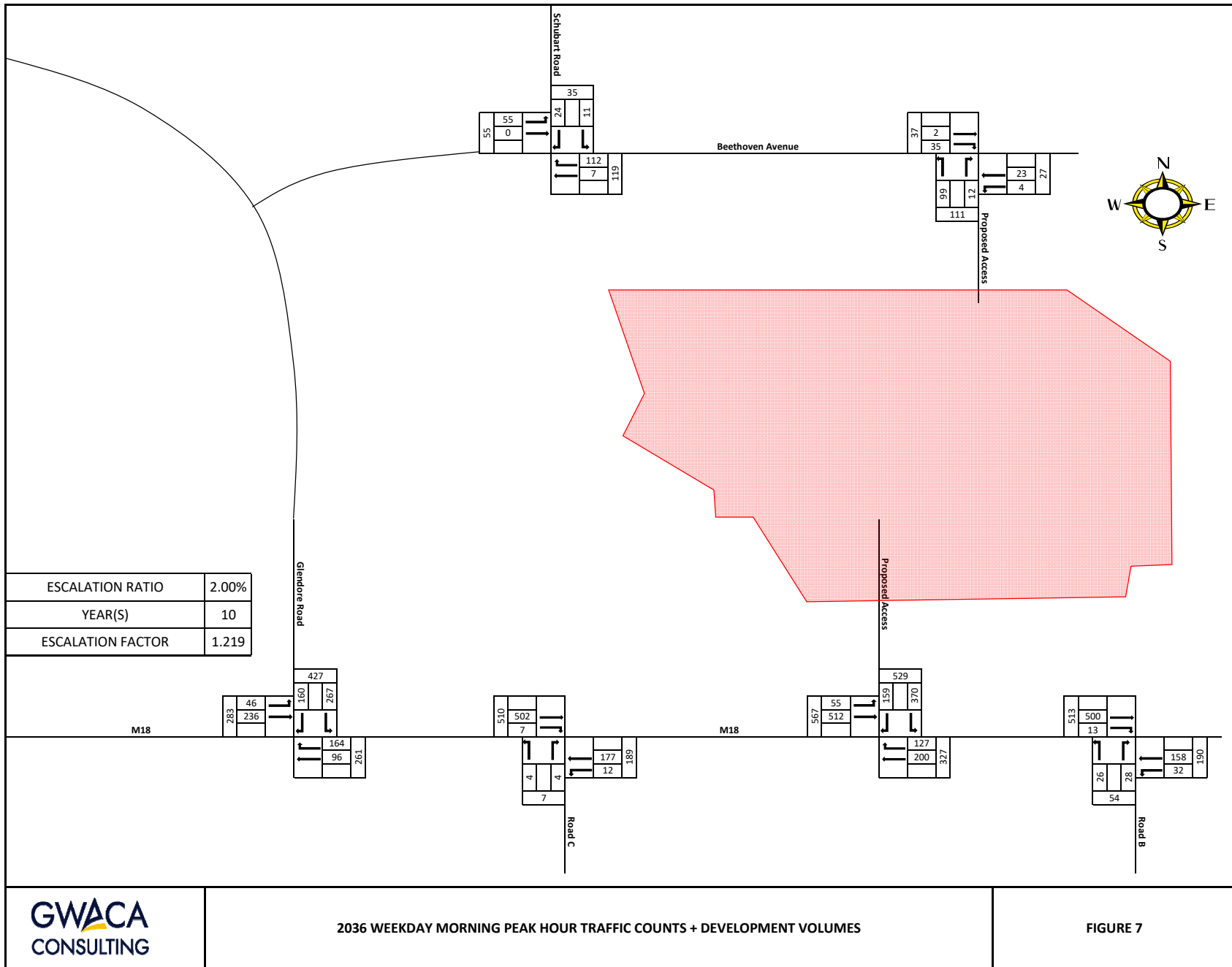


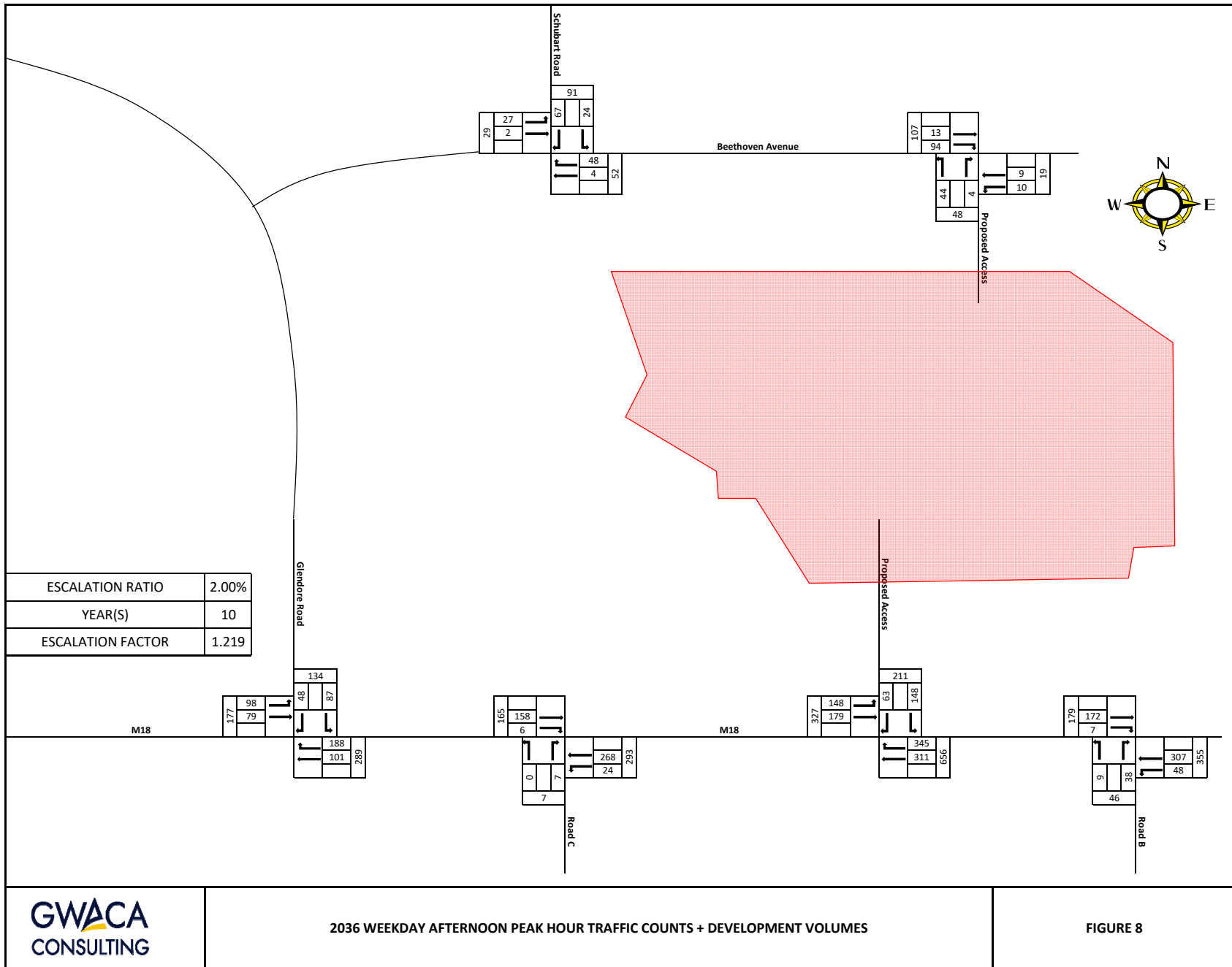












ANNEXURE C: SCENARIO 1 CAPACITY ANALYSIS RESULTS

MOVEMENT SUMMARY



Site: [1] Beethoven Avenue/ Road A (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Veh.]	Dist]									
			veh/h	%	veh/h	%				v/c	sec				veh
South: Road A															
1	L2	All MCs	2	0.0	2	0.0	0.002	8.1	LOS A	0.0	0.1	0.08	0.94	0.08	51.2
3	R2	All MCs	1	0.0	1	0.0	0.002	7.5	LOS A	0.0	0.1	0.08	0.94	0.08	50.9
Approach			3	0.0	3	0.0	0.002	7.9	LOS A	0.0	0.1	0.08	0.94	0.08	51.1
East: Beethoven Avenue															
4	L2	All MCs	1	0.0	1	0.0	0.011	5.5	LOS A	0.0	0.0	0.00	0.03	0.00	57.2
5	T1	All MCs	20	0.0	20	0.0	0.011	0.0	LOS A	0.0	0.0	0.00	0.03	0.00	59.7
Approach			21	0.0	21	0.0	0.011	0.3	NA	0.0	0.0	0.00	0.03	0.00	59.6
West: Beethoven Avenue															
11	T1	All MCs	2	0.0	2	0.0	0.002	0.0	LOS A	0.0	0.1	0.07	0.29	0.07	57.2
12	R2	All MCs	2	0.0	2	0.0	0.002	5.5	LOS A	0.0	0.1	0.07	0.29	0.07	54.6
Approach			4	0.0	4	0.0	0.002	2.8	NA	0.0	0.1	0.07	0.29	0.07	55.9
All Vehicles			28	0.0	28	0.0	0.011	1.5	NA	0.0	0.1	0.02	0.17	0.02	57.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [1] Beethoven Avenue/ Road A (2026)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]		[Total HV]					[Veh. veh	Dist] m				
			veh/h	%	veh/h	%									v/c
South: Road A															
1	L2	All MCs	4	0.0	4	0.0	0.004	8.0	LOS A	0.0	0.1	0.04	0.96	0.04	51.1
3	R2	All MCs	1	0.0	1	0.0	0.004	7.5	LOS A	0.0	0.1	0.04	0.96	0.04	50.9
Approach			5	0.0	5	0.0	0.004	7.9	LOS A	0.0	0.1	0.04	0.96	0.04	51.1
East: Beethoven Avenue															
4	L2	All MCs	1	0.0	1	0.0	0.004	5.5	LOS A	0.0	0.0	0.00	0.07	0.00	56.9
5	T1	All MCs	7	0.0	7	0.0	0.004	0.0	LOS A	0.0	0.0	0.00	0.07	0.00	59.3
Approach			8	0.0	8	0.0	0.004	0.7	NA	0.0	0.0	0.00	0.07	0.00	59.0
West: Beethoven Avenue															
11	T1	All MCs	12	0.0	12	0.0	0.008	0.0	LOS A	0.0	0.1	0.02	0.13	0.02	58.8
12	R2	All MCs	3	0.0	3	0.0	0.008	5.5	LOS A	0.0	0.1	0.02	0.13	0.02	56.1
Approach			15	0.0	15	0.0	0.008	1.2	NA	0.0	0.1	0.02	0.13	0.02	58.2
All Vehicles			28	0.0	28	0.0	0.008	2.3	NA	0.0	0.1	0.02	0.27	0.02	56.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [2] Beethoven Avenue/ Schubert Road (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%				[Veh. veh	Dist]				
			veh/h		veh/h		v/c	sec		veh	m				km/h
East: Beethoven Avenue															
5	T1	All MCs	6	0.0	6	0.0	0.021	0.1	LOS A	0.1	0.7	0.13	0.47	0.13	55.4
6	R2	All MCs	31	0.0	31	0.0	0.021	5.6	LOS A	0.1	0.7	0.13	0.47	0.13	53.0
Approach			37	0.0	37	0.0	0.021	4.6	NA	0.1	0.7	0.13	0.47	0.13	53.4
North: Schubert Road															
7	L2	All MCs	9	0.0	9	0.0	0.026	8.0	LOS A	0.1	0.6	0.03	0.98	0.03	51.3
9	R2	All MCs	21	0.0	21	0.0	0.026	7.5	LOS A	0.1	0.6	0.03	0.98	0.03	51.1
Approach			31	0.0	31	0.0	0.026	7.6	LOS A	0.1	0.6	0.03	0.98	0.03	51.1
West: Beethoven Avenue															
10	L2	All MCs	47	0.0	47	0.0	0.026	5.5	LOS A	0.0	0.0	0.00	0.56	0.00	53.0
11	T1	All MCs	1	0.0	1	0.0	0.026	0.0	LOS A	0.0	0.0	0.00	0.56	0.00	55.1
Approach			48	0.0	48	0.0	0.026	5.4	NA	0.0	0.0	0.00	0.56	0.00	53.0
All Vehicles			116	0.0	116	0.0	0.026	5.8	NA	0.1	0.7	0.05	0.64	0.05	52.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [2] Beethoven Avenue/ Schubert Road (2026)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh.	Dist]				
			[Total HV]	%	[Total HV]	%				veh	m				km/h
East: Beethoven Avenue															
5	T1	All MCs	3	0.0	3	0.0	0.010	0.1	LOS A	0.0	0.3	0.09	0.47	0.09	55.6
6	R2	All MCs	15	0.0	15	0.0	0.010	5.5	LOS A	0.0	0.3	0.09	0.47	0.09	53.2
Approach			18	0.0	18	0.0	0.010	4.5	NA	0.0	0.3	0.09	0.47	0.09	53.6
North: Schubert Road															
7	L2	All MCs	21	0.0	21	0.0	0.066	8.0	LOS A	0.2	1.6	0.04	0.97	0.04	51.3
9	R2	All MCs	58	0.0	58	0.0	0.066	7.5	LOS A	0.2	1.6	0.04	0.97	0.04	51.1
Approach			79	0.0	79	0.0	0.066	7.6	LOS A	0.2	1.6	0.04	0.97	0.04	51.2
West: Beethoven Avenue															
10	L2	All MCs	23	0.0	23	0.0	0.014	5.5	LOS A	0.0	0.0	0.00	0.53	0.00	53.2
11	T1	All MCs	2	0.0	2	0.0	0.014	0.0	LOS A	0.0	0.0	0.00	0.53	0.00	55.4
Approach			25	0.0	25	0.0	0.014	5.1	NA	0.0	0.0	0.00	0.53	0.00	53.4
All Vehicles			122	0.0	122	0.0	0.066	6.6	NA	0.2	1.6	0.04	0.80	0.04	52.0

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [3] M18/ Road B (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	22	0.0	22	0.0	0.058	8.5	LOS A	0.2	1.4	0.37	0.86	0.37	50.2
3	R2	All MCs	24	0.0	24	0.0	0.058	11.1	LOS B	0.2	1.4	0.37	0.86	0.37	50.0
Approach			46	0.0	46	0.0	0.058	9.9	LOS A	0.2	1.4	0.37	0.86	0.37	50.1
East: M18															
4	L2	All MCs	27	0.0	27	0.0	0.085	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
5	T1	All MCs	137	0.0	137	0.0	0.085	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			164	0.0	164	0.0	0.085	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.6
West: M18															
11	T1	All MCs	432	0.0	432	0.0	0.229	0.0	LOS A	0.1	0.6	0.02	0.02	0.02	59.8
12	R2	All MCs	12	0.0	12	0.0	0.229	5.7	LOS A	0.1	0.6	0.02	0.02	0.02	57.0
Approach			443	0.0	443	0.0	0.229	0.2	NA	0.1	0.6	0.02	0.02	0.02	59.7
All Vehicles			654	0.0	654	0.0	0.229	1.0	NA	0.2	1.4	0.04	0.10	0.04	58.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [3] M18/ Road B (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	7	0.0	7	0.0	0.051	9.1	LOS A	0.2	1.2	0.42	0.90	0.42	50.4
3	R2	All MCs	33	0.0	33	0.0	0.051	9.8	LOS A	0.2	1.2	0.42	0.90	0.42	50.2
Approach			40	0.0	40	0.0	0.051	9.6	LOS A	0.2	1.2	0.42	0.90	0.42	50.2
East: M18															
4	L2	All MCs	41	0.0	41	0.0	0.158	5.6	LOS A	0.0	0.0	0.00	0.08	0.00	56.8
5	T1	All MCs	265	0.0	265	0.0	0.158	0.0	LOS A	0.0	0.0	0.00	0.08	0.00	59.2
Approach			306	0.0	306	0.0	0.158	0.8	NA	0.0	0.0	0.00	0.08	0.00	58.9
West: M18															
11	T1	All MCs	148	0.0	148	0.0	0.081	0.1	LOS A	0.1	0.4	0.04	0.05	0.04	59.6
12	R2	All MCs	6	0.0	6	0.0	0.081	6.5	LOS A	0.1	0.4	0.04	0.05	0.04	56.8
Approach			155	0.0	155	0.0	0.081	0.3	NA	0.1	0.4	0.04	0.05	0.04	59.5
All Vehicles			501	0.0	501	0.0	0.158	1.3	NA	0.2	1.2	0.05	0.14	0.05	58.3

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [4] M18/ Road C (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]		[Total HV]					[Veh.	Dist]				
			veh/h	%	veh/h	%	v/c	sec	veh	m			km/h		
South: Road C															
1	L2	All MCs	3	0.0	3	0.0	0.008	8.5	LOS A	0.0	0.2	0.37	0.82	0.37	50.4
3	R2	All MCs	3	0.0	3	0.0	0.008	10.9	LOS B	0.0	0.2	0.37	0.82	0.37	50.1
Approach			6	0.0	6	0.0	0.008	9.7	LOS A	0.0	0.2	0.37	0.82	0.37	50.2
East: M18															
4	L2	All MCs	11	0.0	11	0.0	0.084	5.6	LOS A	0.0	0.0	0.00	0.04	0.00	57.1
5	T1	All MCs	153	0.0	153	0.0	0.084	0.0	LOS A	0.0	0.0	0.00	0.04	0.00	59.6
Approach			163	0.0	163	0.0	0.084	0.4	NA	0.0	0.0	0.00	0.04	0.00	59.4
West: M18															
11	T1	All MCs	434	0.0	434	0.0	0.227	0.0	LOS A	0.1	0.4	0.01	0.01	0.01	59.9
12	R2	All MCs	6	0.0	6	0.0	0.227	5.6	LOS A	0.1	0.4	0.01	0.01	0.01	57.1
Approach			440	0.0	440	0.0	0.227	0.1	NA	0.1	0.4	0.01	0.01	0.01	59.8
All Vehicles			609	0.0	609	0.0	0.227	0.3	NA	0.1	0.4	0.01	0.03	0.01	59.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [4] M18/ Road C (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Veh.]	Dist]									
			veh/h	%						veh/h	%				
South: Road C															
1	L2	All MCs	1	0.0	1	0.0	0.009	8.9	LOS A	0.0	0.2	0.38	0.85	0.38	50.7
3	R2	All MCs	6	0.0	6	0.0	0.009	9.3	LOS A	0.0	0.2	0.38	0.85	0.38	50.4
Approach			7	0.0	7	0.0	0.009	9.2	LOS A	0.0	0.2	0.38	0.85	0.38	50.5
East: M18															
4	L2	All MCs	21	0.0	21	0.0	0.130	5.6	LOS A	0.0	0.0	0.00	0.05	0.00	57.0
5	T1	All MCs	232	0.0	232	0.0	0.130	0.0	LOS A	0.0	0.0	0.00	0.05	0.00	59.5
Approach			253	0.0	253	0.0	0.130	0.5	NA	0.0	0.0	0.00	0.05	0.00	59.3
West: M18															
11	T1	All MCs	137	0.0	137	0.0	0.074	0.0	LOS A	0.0	0.3	0.03	0.04	0.03	59.7
12	R2	All MCs	5	0.0	5	0.0	0.074	6.2	LOS A	0.0	0.3	0.03	0.04	0.03	56.9
Approach			142	0.0	142	0.0	0.074	0.3	NA	0.0	0.3	0.03	0.04	0.03	59.5
All Vehicles			402	0.0	402	0.0	0.130	0.6	NA	0.0	0.3	0.02	0.06	0.02	59.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [5] M18/ Glendore Road (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows										
			[Total HV]	%	[Total HV]	%				[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	83	0.0	83	0.0	0.095	0.6	LOS A	0.4	3.0	0.30	0.37	0.30	56.5
6	R2	All MCs	74	0.0	74	0.0	0.095	6.3	LOS A	0.4	3.0	0.30	0.37	0.30	54.0
Approach			157	0.0	157	0.0	0.095	3.3	NA	0.4	3.0	0.30	0.37	0.30	55.3
North: Glendore Road															
7	L2	All MCs	231	0.0	231	0.0	0.370	9.3	LOS A	1.9	13.0	0.44	0.88	0.46	50.5
9	R2	All MCs	138	0.0	138	0.0	0.370	10.3	LOS B	1.9	13.0	0.44	0.88	0.46	50.3
Approach			368	0.0	368	0.0	0.370	9.7	LOS A	1.9	13.0	0.44	0.88	0.46	50.4
West: M18															
10	L2	All MCs	40	0.0	40	0.0	0.126	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
11	T1	All MCs	204	0.0	204	0.0	0.126	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			244	0.0	244	0.0	0.126	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.6
All Vehicles			769	0.0	769	0.0	0.370	5.6	NA	1.9	13.0	0.27	0.53	0.28	53.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [5] M18/ Glendore Road (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	87	0.0	87	0.0	0.132	0.4	LOS A	0.6	4.5	0.26	0.39	0.26	56.0
6	R2	All MCs	135	0.0	135	0.0	0.132	6.0	LOS A	0.6	4.5	0.26	0.39	0.26	53.5
Approach			222	0.0	222	0.0	0.132	3.8	NA	0.6	4.5	0.26	0.39	0.26	54.5
North: Glendore Road															
7	L2	All MCs	75	0.0	75	0.0	0.105	8.3	LOS A	0.4	2.8	0.22	0.88	0.22	50.9
9	R2	All MCs	41	0.0	41	0.0	0.105	9.2	LOS A	0.4	2.8	0.22	0.88	0.22	50.7
Approach			116	0.0	116	0.0	0.105	8.6	LOS A	0.4	2.8	0.22	0.88	0.22	50.8
West: M18															
10	L2	All MCs	84	0.0	84	0.0	0.080	5.6	LOS A	0.0	0.0	0.00	0.33	0.00	54.8
11	T1	All MCs	68	0.0	68	0.0	0.080	0.0	LOS A	0.0	0.0	0.00	0.33	0.00	57.1
Approach			153	0.0	153	0.0	0.080	3.1	NA	0.0	0.0	0.00	0.33	0.00	55.8
All Vehicles			491	0.0	491	0.0	0.132	4.7	NA	0.6	4.5	0.17	0.49	0.17	54.0

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [6] M18/ Smartstone Access (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	173	0.0	173	0.0	0.091	0.0	LOS A	0.0	0.2	0.02	0.03	0.02	59.8
6	R2	All MCs	3	0.0	3	0.0	0.091	6.9	LOS A	0.0	0.2	0.02	0.03	0.02	57.0
Approach			176	0.0	176	0.0	0.091	0.2	NA	0.0	0.2	0.02	0.03	0.02	59.7
North: Smartstone Access															
7	L2	All MCs	1	0.0	1	0.0	0.003	10.0	LOS A	0.0	0.1	0.48	0.82	0.48	50.0
9	R2	All MCs	1	0.0	1	0.0	0.003	11.0	LOS B	0.0	0.1	0.48	0.82	0.48	49.8
Approach			2	0.0	2	0.0	0.003	10.5	LOS B	0.0	0.1	0.48	0.82	0.48	49.9
West: M18															
10	L2	All MCs	2	0.0	2	0.0	0.228	5.6	LOS A	0.0	0.0	0.00	0.00	0.00	57.4
11	T1	All MCs	442	0.0	442	0.0	0.228	0.1	LOS A	0.0	0.0	0.00	0.00	0.00	59.9
Approach			444	0.0	444	0.0	0.228	0.1	NA	0.0	0.0	0.00	0.00	0.00	59.8
All Vehicles			622	0.0	622	0.0	0.228	0.1	NA	0.0	0.2	0.01	0.01	0.01	59.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [6] M18/ Smartstone Access (2026)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2026 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%				[Veh. veh	Dist]				km/h
East: M18															
5	T1	All MCs	268	0.0	268	0.0	0.138	0.0	LOS A	0.0	0.1	0.00	0.00	0.00	60.0
6	R2	All MCs	1	0.0	1	0.0	0.138	5.5	LOS A	0.0	0.1	0.00	0.00	0.00	57.1
Approach			269	0.0	269	0.0	0.138	0.0	NA	0.0	0.1	0.00	0.00	0.00	60.0
North: Smartstone Access															
7	L2	All MCs	1	0.0	1	0.0	0.002	8.5	LOS A	0.0	0.1	0.32	0.82	0.32	50.8
9	R2	All MCs	1	0.0	1	0.0	0.002	9.5	LOS A	0.0	0.1	0.32	0.82	0.32	50.5
Approach			2	0.0	2	0.0	0.002	9.0	LOS A	0.0	0.1	0.32	0.82	0.32	50.7
West: M18															
10	L2	All MCs	1	0.0	1	0.0	0.080	5.6	LOS A	0.0	0.0	0.00	0.00	0.00	57.4
11	T1	All MCs	155	0.0	155	0.0	0.080	0.0	LOS A	0.0	0.0	0.00	0.00	0.00	59.9
Approach			156	0.0	156	0.0	0.080	0.1	NA	0.0	0.0	0.00	0.00	0.00	59.9
All Vehicles			427	0.0	427	0.0	0.138	0.1	NA	0.0	0.1	0.00	0.01	0.00	59.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

ANNEXURE D: SCENARIO 2 CAPACITY ANALYSIS RESULTS

MOVEMENT SUMMARY



Site: [1.] Beethoven Avenue/ Road A (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Veh.	Dist]									
			veh/h	%	veh/h	%				veh	m				
South: Road A															
1	L2	All MCs	2	0.0	2	0.0	0.002	8.1	LOS A	0.0	0.1	0.08	0.93	0.08	51.2
3	R2	All MCs	1	0.0	1	0.0	0.002	7.5	LOS A	0.0	0.1	0.08	0.93	0.08	50.9
Approach			3	0.0	3	0.0	0.002	7.9	LOS A	0.0	0.1	0.08	0.93	0.08	51.1
East: Beethoven Avenue															
4	L2	All MCs	1	0.0	1	0.0	0.012	5.5	LOS A	0.0	0.0	0.00	0.03	0.00	57.3
5	T1	All MCs	22	0.0	22	0.0	0.012	0.0	LOS A	0.0	0.0	0.00	0.03	0.00	59.7
Approach			23	0.0	23	0.0	0.012	0.3	NA	0.0	0.0	0.00	0.03	0.00	59.6
West: Beethoven Avenue															
11	T1	All MCs	2	0.0	2	0.0	0.002	0.0	LOS A	0.0	0.1	0.07	0.29	0.07	57.2
12	R2	All MCs	2	0.0	2	0.0	0.002	5.5	LOS A	0.0	0.1	0.07	0.29	0.07	54.6
Approach			4	0.0	4	0.0	0.002	2.8	NA	0.0	0.1	0.07	0.29	0.07	55.8
All Vehicles			31	0.0	31	0.0	0.012	1.4	NA	0.0	0.1	0.02	0.16	0.02	58.1

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [1.] Beethoven Avenue/ Road A (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]		[Total HV]					[Veh. veh	Dist] m				
			veh/h	%	veh/h	%									
South: Road A															
1	L2	All MCs	4	0.0	4	0.0	0.004	8.0	LOS A	0.0	0.1	0.05	0.96	0.05	51.1
3	R2	All MCs	1	0.0	1	0.0	0.004	7.5	LOS A	0.0	0.1	0.05	0.96	0.05	50.9
Approach			5	0.0	5	0.0	0.004	7.9	LOS A	0.0	0.1	0.05	0.96	0.05	51.1
East: Beethoven Avenue															
4	L2	All MCs	1	0.0	1	0.0	0.005	5.5	LOS A	0.0	0.0	0.00	0.07	0.00	56.9
5	T1	All MCs	8	0.0	8	0.0	0.005	0.0	LOS A	0.0	0.0	0.00	0.07	0.00	59.4
Approach			9	0.0	9	0.0	0.005	0.6	NA	0.0	0.0	0.00	0.07	0.00	59.1
West: Beethoven Avenue															
11	T1	All MCs	13	0.0	13	0.0	0.008	0.0	LOS A	0.0	0.1	0.02	0.12	0.02	58.9
12	R2	All MCs	3	0.0	3	0.0	0.008	5.5	LOS A	0.0	0.1	0.02	0.12	0.02	56.1
Approach			16	0.0	16	0.0	0.008	1.1	NA	0.0	0.1	0.02	0.12	0.02	58.3
All Vehicles			31	0.0	31	0.0	0.008	2.1	NA	0.0	0.1	0.02	0.25	0.02	57.1

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [2.] Beethoven Avenue/ Schubert Road (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: Beethoven Avenue															
5	T1	All MCs	7	0.0	7	0.0	0.024	0.1	LOS A	0.1	0.8	0.14	0.47	0.14	55.4
6	R2	All MCs	34	0.0	34	0.0	0.024	5.6	LOS A	0.1	0.8	0.14	0.47	0.14	53.0
Approach			41	0.0	41	0.0	0.024	4.6	NA	0.1	0.8	0.14	0.47	0.14	53.4
North: Schubert Road															
7	L2	All MCs	11	0.0	11	0.0	0.029	8.0	LOS A	0.1	0.7	0.03	0.98	0.03	51.3
9	R2	All MCs	23	0.0	23	0.0	0.029	7.5	LOS A	0.1	0.7	0.03	0.98	0.03	51.1
Approach			34	0.0	34	0.0	0.029	7.7	LOS A	0.1	0.7	0.03	0.98	0.03	51.1
West: Beethoven Avenue															
10	L2	All MCs	53	0.0	53	0.0	0.029	5.5	LOS A	0.0	0.0	0.00	0.57	0.00	53.0
11	T1	All MCs	1	0.0	1	0.0	0.029	0.0	LOS A	0.0	0.0	0.00	0.57	0.00	55.1
Approach			54	0.0	54	0.0	0.029	5.4	NA	0.0	0.0	0.00	0.57	0.00	53.0
All Vehicles			128	0.0	128	0.0	0.029	5.8	NA	0.1	0.8	0.05	0.64	0.05	52.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [2.] Beethoven Avenue/ Schubert Road (2031)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV]	%	[Total HV]	%				veh	m				km/h
East: Beethoven Avenue															
5	T1	All MCs	3	0.0	3	0.0	0.011	0.1	LOS A	0.0	0.3	0.09	0.48	0.09	55.5
6	R2	All MCs	16	0.0	16	0.0	0.011	5.5	LOS A	0.0	0.3	0.09	0.48	0.09	53.1
Approach			19	0.0	19	0.0	0.011	4.6	NA	0.0	0.3	0.09	0.48	0.09	53.5
North: Schubert Road															
7	L2	All MCs	23	0.0	23	0.0	0.073	8.0	LOS A	0.3	1.8	0.04	0.97	0.04	51.3
9	R2	All MCs	64	0.0	64	0.0	0.073	7.5	LOS A	0.3	1.8	0.04	0.97	0.04	51.1
Approach			87	0.0	87	0.0	0.073	7.6	LOS A	0.3	1.8	0.04	0.97	0.04	51.2
West: Beethoven Avenue															
10	L2	All MCs	25	0.0	25	0.0	0.015	5.5	LOS A	0.0	0.0	0.00	0.53	0.00	53.2
11	T1	All MCs	2	0.0	2	0.0	0.015	0.0	LOS A	0.0	0.0	0.00	0.53	0.00	55.3
Approach			27	0.0	27	0.0	0.015	5.1	NA	0.0	0.0	0.00	0.53	0.00	53.4
All Vehicles			134	0.0	134	0.0	0.073	6.7	NA	0.3	1.8	0.04	0.81	0.04	51.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [3.] M18/ Road B (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	24	0.0	24	0.0	0.067	8.6	LOS A	0.2	1.6	0.40	0.86	0.40	50.0
3	R2	All MCs	26	0.0	26	0.0	0.067	11.8	LOS B	0.2	1.6	0.40	0.86	0.40	49.8
Approach			51	0.0	51	0.0	0.067	10.3	LOS B	0.2	1.6	0.40	0.86	0.40	49.9
East: M18															
4	L2	All MCs	31	0.0	31	0.0	0.094	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
5	T1	All MCs	152	0.0	152	0.0	0.094	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			182	0.0	182	0.0	0.094	1.0	NA	0.0	0.0	0.00	0.10	0.00	58.6
West: M18															
11	T1	All MCs	477	0.0	477	0.0	0.253	0.0	LOS A	0.1	0.7	0.02	0.02	0.02	59.8
12	R2	All MCs	13	0.0	13	0.0	0.253	5.8	LOS A	0.1	0.7	0.02	0.02	0.02	57.0
Approach			489	0.0	489	0.0	0.253	0.2	NA	0.1	0.7	0.02	0.02	0.02	59.7
All Vehicles			722	0.0	722	0.0	0.253	1.1	NA	0.2	1.6	0.04	0.10	0.04	58.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [3.] M18/ Road B (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	8	0.0	8	0.0	0.059	9.3	LOS A	0.2	1.4	0.44	0.91	0.44	50.3
3	R2	All MCs	36	0.0	36	0.0	0.059	10.1	LOS B	0.2	1.4	0.44	0.91	0.44	50.0
Approach			44	0.0	44	0.0	0.059	10.0	LOS A	0.2	1.4	0.44	0.91	0.44	50.1
East: M18															
4	L2	All MCs	45	0.0	45	0.0	0.174	5.6	LOS A	0.0	0.0	0.00	0.08	0.00	56.8
5	T1	All MCs	293	0.0	293	0.0	0.174	0.0	LOS A	0.0	0.0	0.00	0.08	0.00	59.2
Approach			338	0.0	338	0.0	0.174	0.8	NA	0.0	0.0	0.00	0.08	0.00	58.9
West: M18															
11	T1	All MCs	164	0.0	164	0.0	0.090	0.1	LOS A	0.1	0.4	0.05	0.05	0.05	59.6
12	R2	All MCs	7	0.0	7	0.0	0.090	6.7	LOS A	0.1	0.4	0.05	0.05	0.05	56.8
Approach			172	0.0	172	0.0	0.090	0.4	NA	0.1	0.4	0.05	0.05	0.05	59.4
All Vehicles			554	0.0	554	0.0	0.174	1.4	NA	0.2	1.4	0.05	0.14	0.05	58.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [4.] M18/ Road C (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%		[Veh.]		Dist]					
			veh/h		veh/h	%	v/c	sec		veh	m				
South: Road C															
1	L2	All MCs	3	0.0	3	0.0	0.008	8.6	LOS A	0.0	0.2	0.40	0.82	0.40	50.2
3	R2	All MCs	3	0.0	3	0.0	0.008	11.5	LOS B	0.0	0.2	0.40	0.82	0.40	49.9
Approach			6	0.0	6	0.0	0.008	10.0	LOS B	0.0	0.2	0.40	0.82	0.40	50.0
East: M18															
4	L2	All MCs	12	0.0	12	0.0	0.093	5.6	LOS A	0.0	0.0	0.00	0.04	0.00	57.1
5	T1	All MCs	168	0.0	168	0.0	0.093	0.0	LOS A	0.0	0.0	0.00	0.04	0.00	59.6
Approach			180	0.0	180	0.0	0.093	0.4	NA	0.0	0.0	0.00	0.04	0.00	59.4
West: M18															
11	T1	All MCs	479	0.0	479	0.0	0.251	0.0	LOS A	0.1	0.4	0.01	0.01	0.01	59.9
12	R2	All MCs	7	0.0	7	0.0	0.251	5.6	LOS A	0.1	0.4	0.01	0.01	0.01	57.0
Approach			486	0.0	486	0.0	0.251	0.1	NA	0.1	0.4	0.01	0.01	0.01	59.8
All Vehicles			673	0.0	673	0.0	0.251	0.3	NA	0.1	0.4	0.01	0.03	0.01	59.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [4.] M18/ Road C (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Cycles Rate to Depart	Number of Cycles	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh. veh	Dist] m				km/h
			veh/h	%	veh/h	%	v/c	sec							
South: Road C															
1	L2	All MCs	1	0.0	1	0.0	0.011	9.0	LOS A	0.0	0.2	0.41	0.86	0.41	50.5
3	R2	All MCs	7	0.0	7	0.0	0.011	9.6	LOS A	0.0	0.2	0.41	0.86	0.41	50.3
Approach			8	0.0	8	0.0	0.011	9.5	LOS A	0.0	0.2	0.41	0.86	0.41	50.3
East: M18															
4	L2	All MCs	23	0.0	23	0.0	0.144	5.6	LOS A	0.0	0.0	0.00	0.05	0.00	57.0
5	T1	All MCs	256	0.0	256	0.0	0.144	0.0	LOS A	0.0	0.0	0.00	0.05	0.00	59.5
Approach			279	0.0	279	0.0	0.144	0.5	NA	0.0	0.0	0.00	0.05	0.00	59.3
West: M18															
11	T1	All MCs	164	0.0	164	0.0	0.090	0.1	LOS A	0.1	0.4	0.04	0.05	0.04	59.6
12	R2	All MCs	7	0.0	7	0.0	0.090	6.4	LOS A	0.1	0.4	0.04	0.05	0.04	56.8
Approach			172	0.0	172	0.0	0.090	0.3	NA	0.1	0.4	0.04	0.05	0.04	59.5
All Vehicles			459	0.0	459	0.0	0.144	0.6	NA	0.1	0.4	0.02	0.06	0.02	59.1

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [5.] M18/ Glendore Road (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%				[Veh.]	Dist]				
			veh/h		veh/h	%				v/c	sec				veh
East: M18															
5	T1	All MCs	92	0.0	92	0.0	0.106	0.7	LOS A	0.5	3.4	0.32	0.38	0.32	56.4
6	R2	All MCs	81	0.0	81	0.0	0.106	6.4	LOS A	0.5	3.4	0.32	0.38	0.32	53.9
Approach			173	0.0	173	0.0	0.106	3.4	NA	0.5	3.4	0.32	0.38	0.32	55.2
North: Glendore Road															
7	L2	All MCs	255	0.0	255	0.0	0.423	9.8	LOS A	2.5	17.8	0.49	0.90	0.57	50.1
9	R2	All MCs	153	0.0	153	0.0	0.423	11.2	LOS B	2.5	17.8	0.49	0.90	0.57	49.9
Approach			407	0.0	407	0.0	0.423	10.3	LOS B	2.5	17.8	0.49	0.90	0.57	50.0
West: M18															
10	L2	All MCs	44	0.0	44	0.0	0.139	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
11	T1	All MCs	225	0.0	225	0.0	0.139	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			269	0.0	269	0.0	0.139	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.6
All Vehicles			849	0.0	849	0.0	0.423	5.9	NA	2.5	17.8	0.30	0.54	0.34	53.5

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [5.] M18/ Glendore Road (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	97	0.0	97	0.0	0.146	0.5	LOS A	0.7	5.1	0.28	0.40	0.28	56.0
6	R2	All MCs	148	0.0	148	0.0	0.146	6.0	LOS A	0.7	5.1	0.28	0.40	0.28	53.5
Approach			245	0.0	245	0.0	0.146	3.8	NA	0.7	5.1	0.28	0.40	0.28	54.4
North: Glendore Road															
7	L2	All MCs	82	0.0	82	0.0	0.118	8.3	LOS A	0.5	3.2	0.23	0.88	0.23	50.9
9	R2	All MCs	45	0.0	45	0.0	0.118	9.5	LOS A	0.5	3.2	0.23	0.88	0.23	50.6
Approach			127	0.0	127	0.0	0.118	8.7	LOS A	0.5	3.2	0.23	0.88	0.23	50.8
West: M18															
10	L2	All MCs	86	0.0	86	0.0	0.085	5.6	LOS A	0.0	0.0	0.00	0.31	0.00	54.9
11	T1	All MCs	76	0.0	76	0.0	0.085	0.0	LOS A	0.0	0.0	0.00	0.31	0.00	57.2
Approach			162	0.0	162	0.0	0.085	3.0	NA	0.0	0.0	0.00	0.31	0.00	56.0
All Vehicles			535	0.0	535	0.0	0.146	4.7	NA	0.7	5.1	0.18	0.49	0.18	54.0

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [6.] M18/ Smartstone Access (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows										
			[Total HV]	%	[Total HV]	%				veh/h					[Veh.
East: M18															
5	T1	All MCs	191	0.0	191	0.0	0.101	0.0	LOS A	0.0	0.2	0.02	0.03	0.02	59.8
6	R2	All MCs	3	0.0	3	0.0	0.101	7.2	LOS A	0.0	0.2	0.02	0.03	0.02	57.0
Approach			194	0.0	194	0.0	0.101	0.2	NA	0.0	0.2	0.02	0.03	0.02	59.8
North: Smartstone Access															
7	L2	All MCs	1	0.0	1	0.0	0.003	10.3	LOS B	0.0	0.1	0.51	0.83	0.51	49.7
9	R2	All MCs	1	0.0	1	0.0	0.003	11.6	LOS B	0.0	0.1	0.51	0.83	0.51	49.5
Approach			2	0.0	2	0.0	0.003	10.9	LOS B	0.0	0.1	0.51	0.83	0.51	49.6
West: M18															
10	L2	All MCs	2	0.0	2	0.0	0.252	5.6	LOS A	0.0	0.0	0.00	0.00	0.00	57.4
11	T1	All MCs	488	0.0	488	0.0	0.252	0.1	LOS A	0.0	0.0	0.00	0.00	0.00	59.8
Approach			491	0.0	491	0.0	0.252	0.1	NA	0.0	0.0	0.00	0.00	0.00	59.8
All Vehicles			686	0.0	686	0.0	0.252	0.1	NA	0.0	0.2	0.01	0.01	0.01	59.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [6.] M18/ Smartstone Access (2031)

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows										
			[Total HV]	%	[Total HV]	%				veh	m				km/h
East: M18															
5	T1	All MCs	297	0.0	297	0.0	0.153	0.0	LOS A	0.0	0.1	0.00	0.00	0.00	60.0
6	R2	All MCs	1	0.0	1	0.0	0.153	5.5	LOS A	0.0	0.1	0.00	0.00	0.00	57.1
Approach			298	0.0	298	0.0	0.153	0.0	NA	0.0	0.1	0.00	0.00	0.00	60.0
North: Smartstone Access															
7	L2	All MCs	1	0.0	1	0.0	0.002	8.6	LOS A	0.0	0.1	0.34	0.81	0.34	50.7
9	R2	All MCs	1	0.0	1	0.0	0.002	9.7	LOS A	0.0	0.1	0.34	0.81	0.34	50.5
Approach			2	0.0	2	0.0	0.002	9.2	LOS A	0.0	0.1	0.34	0.81	0.34	50.6
West: M18															
10	L2	All MCs	1	0.0	1	0.0	0.088	5.6	LOS A	0.0	0.0	0.00	0.00	0.00	57.4
11	T1	All MCs	171	0.0	171	0.0	0.088	0.0	LOS A	0.0	0.0	0.00	0.00	0.00	59.9
Approach			172	0.0	172	0.0	0.088	0.1	NA	0.0	0.0	0.00	0.00	0.00	59.9
All Vehicles			472	0.0	472	0.0	0.153	0.1	NA	0.0	0.1	0.00	0.01	0.00	59.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

ANNEXURE E: SCENARIO 3 CAPACITY ANALYSIS RESULTS

MOVEMENT SUMMARY



Site: [.1] Beethoven Avenue/ Road A (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh.]	[Dist]				
			veh/h	%	veh/h	%	v/c	sec		veh	m				km/h
South: Road A															
1	L2	All MCs	104	0.0	104	0.0	0.085	8.1	LOS A	0.3	2.4	0.09	0.93	0.09	51.1
3	R2	All MCs	13	0.0	13	0.0	0.085	7.7	LOS A	0.3	2.4	0.09	0.93	0.09	50.9
Approach			117	0.0	117	0.0	0.085	8.1	LOS A	0.3	2.4	0.09	0.93	0.09	51.1
East: Beethoven Avenue															
4	L2	All MCs	4	0.0	4	0.0	0.014	5.5	LOS A	0.0	0.0	0.00	0.10	0.00	56.7
5	T1	All MCs	22	0.0	22	0.0	0.014	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			26	0.0	26	0.0	0.014	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.7
West: Beethoven Avenue															
11	T1	All MCs	2	0.0	2	0.0	0.022	0.1	LOS A	0.1	0.7	0.09	0.54	0.09	55.0
12	R2	All MCs	37	0.0	37	0.0	0.022	5.5	LOS A	0.1	0.7	0.09	0.54	0.09	52.6
Approach			39	0.0	39	0.0	0.022	5.2	NA	0.1	0.7	0.09	0.54	0.09	52.8
All Vehicles			182	0.0	182	0.0	0.085	6.4	NA	0.3	2.4	0.08	0.73	0.08	52.4

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [1] Beethoven Avenue/ Road A (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
South: Road A															
1	L2	All MCs	45	0.0	45	0.0	0.036	8.0	LOS A	0.1	1.0	0.05	0.96	0.05	51.1
3	R2	All MCs	4	0.0	4	0.0	0.036	7.7	LOS A	0.1	1.0	0.05	0.96	0.05	50.8
Approach			49	0.0	49	0.0	0.036	8.0	LOS A	0.1	1.0	0.05	0.96	0.05	51.1
East: Beethoven Avenue															
4	L2	All MCs	11	0.0	11	0.0	0.010	5.5	LOS A	0.0	0.0	0.00	0.33	0.00	54.8
5	T1	All MCs	8	0.0	8	0.0	0.010	0.0	LOS A	0.0	0.0	0.00	0.33	0.00	57.1
Approach			19	0.0	19	0.0	0.010	3.1	NA	0.0	0.0	0.00	0.33	0.00	55.8
West: Beethoven Avenue															
11	T1	All MCs	13	0.0	13	0.0	0.062	0.0	LOS A	0.3	2.1	0.08	0.51	0.08	55.3
12	R2	All MCs	98	0.0	98	0.0	0.062	5.5	LOS A	0.3	2.1	0.08	0.51	0.08	52.9
Approach			111	0.0	111	0.0	0.062	4.9	NA	0.3	2.1	0.08	0.51	0.08	53.2
All Vehicles			179	0.0	179	0.0	0.062	5.5	NA	0.3	2.1	0.06	0.61	0.06	52.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [.2] Beethoven Avenue/ Schubert Road (2031 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				v/c	sec				veh
East: Beethoven Avenue															
5	T1	All MCs	7	0.0	7	0.0	0.071	0.2	LOS A	0.3	2.3	0.14	0.53	0.14	54.9
6	R2	All MCs	115	0.0	115	0.0	0.071	5.6	LOS A	0.3	2.3	0.14	0.53	0.14	52.5
Approach			122	0.0	122	0.0	0.071	5.3	NA	0.3	2.3	0.14	0.53	0.14	52.7
North: Schubert Road															
7	L2	All MCs	11	0.0	11	0.0	0.030	8.0	LOS A	0.1	0.7	0.03	0.97	0.03	51.3
9	R2	All MCs	23	0.0	23	0.0	0.030	7.5	LOS A	0.1	0.7	0.03	0.97	0.03	51.1
Approach			34	0.0	34	0.0	0.030	7.7	LOS A	0.1	0.7	0.03	0.97	0.03	51.1
West: Beethoven Avenue															
10	L2	All MCs	53	0.0	53	0.0	0.029	5.5	LOS A	0.0	0.0	0.00	0.57	0.00	53.0
11	T1	All MCs	1	0.0	1	0.0	0.029	0.0	LOS A	0.0	0.0	0.00	0.57	0.00	55.1
Approach			54	0.0	54	0.0	0.029	5.4	NA	0.0	0.0	0.00	0.57	0.00	53.0
All Vehicles			209	0.0	209	0.0	0.071	5.7	NA	0.3	2.3	0.09	0.61	0.09	52.5

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [.2] Beethoven Avenue/ Schubert Road (2031 + Development)**

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg.	Aver.	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows		Satn	Delay		[Veh. veh	Dist] m				
			[Total HV] veh/h	%	[Total HV] veh/h	%	v/c	sec							
East: Beethoven Avenue															
5	T1	All MCs	3	0.0	3	0.0	0.030	0.1	LOS A	0.1	1.0	0.09	0.53	0.09	55.0
6	R2	All MCs	49	0.0	49	0.0	0.030	5.5	LOS A	0.1	1.0	0.09	0.53	0.09	52.7
Approach			53	0.0	53	0.0	0.030	5.2	NA	0.1	1.0	0.09	0.53	0.09	52.8
North: Schubert Road															
7	L2	All MCs	23	0.0	23	0.0	0.075	8.0	LOS A	0.3	1.9	0.04	0.96	0.04	51.3
9	R2	All MCs	64	0.0	64	0.0	0.075	7.5	LOS A	0.3	1.9	0.04	0.96	0.04	51.1
Approach			87	0.0	87	0.0	0.075	7.7	LOS A	0.3	1.9	0.04	0.96	0.04	51.2
West: Beethoven Avenue															
10	L2	All MCs	25	0.0	25	0.0	0.015	5.5	LOS A	0.0	0.0	0.00	0.53	0.00	53.2
11	T1	All MCs	2	0.0	2	0.0	0.015	0.0	LOS A	0.0	0.0	0.00	0.53	0.00	55.3
Approach			27	0.0	27	0.0	0.015	5.1	NA	0.0	0.0	0.00	0.53	0.00	53.4
All Vehicles			167	0.0	167	0.0	0.075	6.5	NA	0.3	1.9	0.05	0.76	0.05	52.0

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.3] M18/ Road B (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	24	0.0	24	0.0	0.067	8.6	LOS A	0.2	1.6	0.40	0.86	0.40	50.0
3	R2	All MCs	26	0.0	26	0.0	0.067	11.8	LOS B	0.2	1.6	0.40	0.86	0.40	49.8
Approach			51	0.0	51	0.0	0.067	10.3	LOS B	0.2	1.6	0.40	0.86	0.40	49.9
East: M18															
4	L2	All MCs	31	0.0	31	0.0	0.094	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
5	T1	All MCs	152	0.0	152	0.0	0.094	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			182	0.0	182	0.0	0.094	1.0	NA	0.0	0.0	0.00	0.10	0.00	58.6
West: M18															
11	T1	All MCs	477	0.0	477	0.0	0.253	0.0	LOS A	0.1	0.7	0.02	0.02	0.02	59.8
12	R2	All MCs	13	0.0	13	0.0	0.253	5.8	LOS A	0.1	0.7	0.02	0.02	0.02	57.0
Approach			489	0.0	489	0.0	0.253	0.2	NA	0.1	0.7	0.02	0.02	0.02	59.7
All Vehicles			722	0.0	722	0.0	0.253	1.1	NA	0.2	1.6	0.04	0.10	0.04	58.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.3] M18/ Road B (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				v/c	sec				veh
South: Road B															
1	L2	All MCs	8	0.0	8	0.0	0.059	9.3	LOS A	0.2	1.4	0.44	0.91	0.44	50.3
3	R2	All MCs	36	0.0	36	0.0	0.059	10.1	LOS B	0.2	1.4	0.44	0.91	0.44	50.0
Approach			44	0.0	44	0.0	0.059	10.0	LOS A	0.2	1.4	0.44	0.91	0.44	50.1
East: M18															
4	L2	All MCs	45	0.0	45	0.0	0.174	5.6	LOS A	0.0	0.0	0.00	0.08	0.00	56.8
5	T1	All MCs	293	0.0	293	0.0	0.174	0.0	LOS A	0.0	0.0	0.00	0.08	0.00	59.2
Approach			338	0.0	338	0.0	0.174	0.8	NA	0.0	0.0	0.00	0.08	0.00	58.9
West: M18															
11	T1	All MCs	164	0.0	164	0.0	0.090	0.1	LOS A	0.1	0.4	0.05	0.05	0.05	59.6
12	R2	All MCs	7	0.0	7	0.0	0.090	6.7	LOS A	0.1	0.4	0.05	0.05	0.05	56.8
Approach			172	0.0	172	0.0	0.090	0.4	NA	0.1	0.4	0.05	0.05	0.05	59.4
All Vehicles			554	0.0	554	0.0	0.174	1.4	NA	0.2	1.4	0.05	0.14	0.05	58.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.4] M18/ Road C (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh. veh	Dist] m				km/h
			veh/h	%	veh/h	%	v/c	sec							
South: Road C															
1	L2	All MCs	3	0.0	3	0.0	0.008	8.6	LOS A	0.0	0.2	0.40	0.82	0.40	50.2
3	R2	All MCs	3	0.0	3	0.0	0.008	11.5	LOS B	0.0	0.2	0.40	0.82	0.40	49.9
Approach			6	0.0	6	0.0	0.008	10.0	LOS B	0.0	0.2	0.40	0.82	0.40	50.0
East: M18															
4	L2	All MCs	12	0.0	12	0.0	0.093	5.6	LOS A	0.0	0.0	0.00	0.04	0.00	57.1
5	T1	All MCs	168	0.0	168	0.0	0.093	0.0	LOS A	0.0	0.0	0.00	0.04	0.00	59.6
Approach			180	0.0	180	0.0	0.093	0.4	NA	0.0	0.0	0.00	0.04	0.00	59.4
West: M18															
11	T1	All MCs	479	0.0	479	0.0	0.251	0.0	LOS A	0.1	0.4	0.01	0.01	0.01	59.9
12	R2	All MCs	7	0.0	7	0.0	0.251	5.6	LOS A	0.1	0.4	0.01	0.01	0.01	57.0
Approach			486	0.0	486	0.0	0.251	0.1	NA	0.1	0.4	0.01	0.01	0.01	59.8
All Vehicles			673	0.0	673	0.0	0.251	0.3	NA	0.1	0.4	0.01	0.03	0.01	59.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.4] M18/ Road C (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh.]	[Dist]				
			veh/h	%	veh/h	%	v/c	sec		veh	m				km/h
South: Road C															
1	L2	All MCs	1	0.0	1	0.0	0.011	9.0	LOS A	0.0	0.2	0.41	0.86	0.41	50.5
3	R2	All MCs	7	0.0	7	0.0	0.011	9.5	LOS A	0.0	0.2	0.41	0.86	0.41	50.3
Approach			8	0.0	8	0.0	0.011	9.5	LOS A	0.0	0.2	0.41	0.86	0.41	50.3
East: M18															
4	L2	All MCs	23	0.0	23	0.0	0.144	5.6	LOS A	0.0	0.0	0.00	0.05	0.00	57.0
5	T1	All MCs	256	0.0	256	0.0	0.144	0.0	LOS A	0.0	0.0	0.00	0.05	0.00	59.5
Approach			279	0.0	279	0.0	0.144	0.5	NA	0.0	0.0	0.00	0.05	0.00	59.3
West: M18															
11	T1	All MCs	152	0.0	152	0.0	0.083	0.1	LOS A	0.1	0.4	0.04	0.05	0.04	59.6
12	R2	All MCs	7	0.0	7	0.0	0.083	6.4	LOS A	0.1	0.4	0.04	0.05	0.04	56.8
Approach			159	0.0	159	0.0	0.083	0.4	NA	0.1	0.4	0.04	0.05	0.04	59.4
All Vehicles			446	0.0	446	0.0	0.144	0.6	NA	0.1	0.4	0.02	0.07	0.02	59.1

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [.5] M18/ Glendore Road (2031 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					Veh.	Dist				
			[Total HV]	%	[Total HV]	%				veh	m				km/h
East: M18															
5	T1	All MCs	92	0.0	92	0.0	0.165	0.9	LOS A	0.8	5.9	0.38	0.48	0.38	55.5
6	R2	All MCs	164	0.0	164	0.0	0.165	6.4	LOS A	0.8	5.9	0.38	0.48	0.38	53.0
Approach			256	0.0	256	0.0	0.165	4.5	NA	0.8	5.9	0.38	0.48	0.38	53.9
North: Glendore Road															
7	L2	All MCs	255	0.0	255	0.0	0.443	10.0	LOS A	2.8	19.8	0.51	0.92	0.63	49.8
9	R2	All MCs	153	0.0	153	0.0	0.443	12.3	LOS B	2.8	19.8	0.51	0.92	0.63	49.5
Approach			407	0.0	407	0.0	0.443	10.9	LOS B	2.8	19.8	0.51	0.92	0.63	49.7
West: M18															
10	L2	All MCs	44	0.0	44	0.0	0.139	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
11	T1	All MCs	225	0.0	225	0.0	0.139	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.1
Approach			269	0.0	269	0.0	0.139	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.6
All Vehicles			933	0.0	933	0.0	0.443	6.2	NA	2.8	19.8	0.33	0.56	0.38	53.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.5] M18/ Glendore Road (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. veh	Dist] m				
			[Total HV] veh/h	[Total HV] %	[Total HV] veh/h	[Total HV] %									
East: M18															
5	T1	All MCs	97	0.0	97	0.0	0.168	0.5	LOS A	0.9	6.0	0.29	0.42	0.29	55.7
6	R2	All MCs	182	0.0	182	0.0	0.168	6.0	LOS A	0.9	6.0	0.29	0.42	0.29	53.3
Approach			279	0.0	279	0.0	0.168	4.1	NA	0.9	6.0	0.29	0.42	0.29	54.1
North: Glendore Road															
7	L2	All MCs	82	0.0	82	0.0	0.120	8.3	LOS A	0.5	3.3	0.24	0.88	0.24	50.8
9	R2	All MCs	45	0.0	45	0.0	0.120	9.7	LOS A	0.5	3.3	0.24	0.88	0.24	50.6
Approach			127	0.0	127	0.0	0.120	8.8	LOS A	0.5	3.3	0.24	0.88	0.24	50.7
West: M18															
10	L2	All MCs	86	0.0	86	0.0	0.085	5.6	LOS A	0.0	0.0	0.00	0.31	0.00	54.9
11	T1	All MCs	76	0.0	76	0.0	0.085	0.0	LOS A	0.0	0.0	0.00	0.31	0.00	57.2
Approach			162	0.0	162	0.0	0.085	3.0	NA	0.0	0.0	0.00	0.31	0.00	56.0
All Vehicles			568	0.0	568	0.0	0.168	4.8	NA	0.9	6.0	0.19	0.49	0.19	53.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [.6] M18/ Proposed Access (2031 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				veh	m				
East: M18															
5	T1	All MCs	191	0.0	191	0.0	0.231	1.8	LOS A	1.2	8.2	0.47	0.55	0.47	56.0
6	R2	All MCs	134	0.0	134	0.0	0.231	8.0	LOS A	1.2	8.2	0.47	0.55	0.47	53.5
Approach			324	0.0	324	0.0	0.231	4.4	NA	1.2	8.2	0.47	0.55	0.47	54.9
North: Proposed Access															
7	L2	All MCs	389	0.0	389	0.0	0.858	21.0	LOS C	10.8	75.8	0.87	1.53	2.45	43.1
9	R2	All MCs	167	0.0	167	0.0	0.858	27.8	LOS D	10.8	75.8	0.87	1.53	2.45	42.9
Approach			557	0.0	557	0.0	0.858	23.0	LOS C	10.8	75.8	0.87	1.53	2.45	43.0
West: M18															
10	L2	All MCs	58	0.0	58	0.0	0.282	5.6	LOS A	0.0	0.0	0.00	0.06	0.00	56.8
11	T1	All MCs	488	0.0	488	0.0	0.282	0.1	LOS A	0.0	0.0	0.00	0.06	0.00	59.3
Approach			546	0.0	546	0.0	0.282	0.7	NA	0.0	0.0	0.00	0.06	0.00	59.0
All Vehicles			1427	0.0	1427	0.0	0.858	10.2	NA	10.8	75.8	0.45	0.74	1.06	50.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [.6] M18/ Proposed Access (2031 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2031 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	297	0.0	297	0.0	0.431	1.6	LOS A	3.1	21.4	0.48	0.55	0.53	55.4
6	R2	All MCs	363	0.0	363	0.0	0.431	7.3	LOS A	3.1	21.4	0.48	0.55	0.53	53.0
Approach			660	0.0	660	0.0	0.431	4.8	NA	3.1	21.4	0.48	0.55	0.53	54.1
North: Proposed Access															
7	L2	All MCs	156	0.0	156	0.0	0.306	8.9	LOS A	1.3	9.0	0.50	0.83	0.50	49.3
9	R2	All MCs	66	0.0	66	0.0	0.306	17.7	LOS C	1.3	9.0	0.50	0.83	0.50	49.1
Approach			222	0.0	222	0.0	0.306	11.5	LOS B	1.3	9.0	0.50	0.83	0.50	49.3
West: M18															
10	L2	All MCs	156	0.0	156	0.0	0.171	5.6	LOS A	0.0	0.0	0.00	0.28	0.00	55.1
11	T1	All MCs	171	0.0	171	0.0	0.171	0.0	LOS A	0.0	0.0	0.00	0.28	0.00	57.4
Approach			326	0.0	326	0.0	0.171	2.7	NA	0.0	0.0	0.00	0.28	0.00	56.3
All Vehicles			1208	0.0	1208	0.0	0.431	5.4	NA	3.1	21.4	0.35	0.53	0.38	53.7

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

ANNEXURE F: SCENARIO 4 CAPACITY ANALYSIS RESULTS

MOVEMENT SUMMARY



Site: [1] Beethoven Avenue/ Road A (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Veh.]	Dist]									
			veh/h	%	veh/h	%				v/c	sec				veh
South: Road A															
1	L2	All MCs	104	0.0	104	0.0	0.086	8.1	LOS A	0.3	2.4	0.09	0.93	0.09	51.1
3	R2	All MCs	13	0.0	13	0.0	0.086	7.7	LOS A	0.3	2.4	0.09	0.93	0.09	50.9
Approach			117	0.0	117	0.0	0.086	8.1	LOS A	0.3	2.4	0.09	0.93	0.09	51.1
East: Beethoven Avenue															
4	L2	All MCs	4	0.0	4	0.0	0.015	5.5	LOS A	0.0	0.0	0.00	0.09	0.00	56.8
5	T1	All MCs	24	0.0	24	0.0	0.015	0.0	LOS A	0.0	0.0	0.00	0.09	0.00	59.2
Approach			28	0.0	28	0.0	0.015	0.8	NA	0.0	0.0	0.00	0.09	0.00	58.8
West: Beethoven Avenue															
11	T1	All MCs	2	0.0	2	0.0	0.022	0.1	LOS A	0.1	0.7	0.09	0.54	0.09	55.0
12	R2	All MCs	37	0.0	37	0.0	0.022	5.5	LOS A	0.1	0.7	0.09	0.54	0.09	52.6
Approach			39	0.0	39	0.0	0.022	5.2	NA	0.1	0.7	0.09	0.54	0.09	52.7
All Vehicles			184	0.0	184	0.0	0.086	6.3	NA	0.3	2.4	0.08	0.72	0.08	52.5

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [1.] Beethoven Avenue/ Road A (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Veh. veh	Dist] m									
			veh/h	%	veh/h	%				v/c	sec				veh
South: Road A															
1	L2	All MCs	46	0.0	46	0.0	0.037	8.0	LOS A	0.1	1.0	0.05	0.96	0.05	51.1
3	R2	All MCs	4	0.0	4	0.0	0.037	7.7	LOS A	0.1	1.0	0.05	0.96	0.05	50.8
Approach			51	0.0	51	0.0	0.037	8.0	LOS A	0.1	1.0	0.05	0.96	0.05	51.1
East: Beethoven Avenue															
4	L2	All MCs	11	0.0	11	0.0	0.011	5.5	LOS A	0.0	0.0	0.00	0.31	0.00	55.0
5	T1	All MCs	9	0.0	9	0.0	0.011	0.0	LOS A	0.0	0.0	0.00	0.31	0.00	57.2
Approach			20	0.0	20	0.0	0.011	2.9	NA	0.0	0.0	0.00	0.31	0.00	56.0
West: Beethoven Avenue															
11	T1	All MCs	14	0.0	14	0.0	0.063	0.0	LOS A	0.3	2.1	0.08	0.50	0.08	55.4
12	R2	All MCs	98	0.0	98	0.0	0.063	5.5	LOS A	0.3	2.1	0.08	0.50	0.08	53.0
Approach			112	0.0	112	0.0	0.063	4.8	NA	0.3	2.1	0.08	0.50	0.08	53.2
All Vehicles			182	0.0	182	0.0	0.063	5.5	NA	0.3	2.1	0.06	0.61	0.06	52.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [2] Beethoven Avenue/ Schubert Road (2036 + Development)**

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				v/c	sec				veh
East: Beethoven Avenue															
5	T1	All MCs	7	0.0	7	0.0	0.073	0.2	LOS A	0.3	2.4	0.15	0.53	0.15	54.9
6	R2	All MCs	118	0.0	118	0.0	0.073	5.6	LOS A	0.3	2.4	0.15	0.53	0.15	52.5
Approach			125	0.0	125	0.0	0.073	5.3	NA	0.3	2.4	0.15	0.53	0.15	52.6
North: Schubert Road															
7	L2	All MCs	12	0.0	12	0.0	0.033	8.0	LOS A	0.1	0.8	0.03	0.97	0.03	51.3
9	R2	All MCs	25	0.0	25	0.0	0.033	7.6	LOS A	0.1	0.8	0.03	0.97	0.03	51.0
Approach			37	0.0	37	0.0	0.033	7.7	LOS A	0.1	0.8	0.03	0.97	0.03	51.1
West: Beethoven Avenue															
10	L2	All MCs	58	0.0	58	0.0	0.032	5.5	LOS A	0.0	0.0	0.00	0.57	0.00	53.0
11	T1	All MCs	1	0.0	1	0.0	0.032	0.0	LOS A	0.0	0.0	0.00	0.57	0.00	55.1
Approach			59	0.0	59	0.0	0.032	5.4	NA	0.0	0.0	0.00	0.57	0.00	53.0
All Vehicles			221	0.0	221	0.0	0.073	5.7	NA	0.3	2.4	0.09	0.61	0.09	52.5

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [2.] Beethoven Avenue/ Schubert Road (2036 + Development)**

Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				v/c	sec				veh
East: Beethoven Avenue															
5	T1	All MCs	4	0.0	4	0.0	0.031	0.1	LOS A	0.1	1.0	0.10	0.52	0.10	55.1
6	R2	All MCs	51	0.0	51	0.0	0.031	5.5	LOS A	0.1	1.0	0.10	0.52	0.10	52.7
Approach			55	0.0	55	0.0	0.031	5.1	NA	0.1	1.0	0.10	0.52	0.10	52.9
North: Schubert Road															
7	L2	All MCs	25	0.0	25	0.0	0.082	8.0	LOS A	0.3	2.1	0.04	0.96	0.04	51.3
9	R2	All MCs	71	0.0	71	0.0	0.082	7.6	LOS A	0.3	2.1	0.04	0.96	0.04	51.1
Approach			96	0.0	96	0.0	0.082	7.7	LOS A	0.3	2.1	0.04	0.96	0.04	51.2
West: Beethoven Avenue															
10	L2	All MCs	28	0.0	28	0.0	0.016	5.5	LOS A	0.0	0.0	0.00	0.54	0.00	53.2
11	T1	All MCs	2	0.0	2	0.0	0.016	0.0	LOS A	0.0	0.0	0.00	0.54	0.00	55.3
Approach			31	0.0	31	0.0	0.016	5.2	NA	0.0	0.0	0.00	0.54	0.00	53.3
All Vehicles			181	0.0	181	0.0	0.082	6.5	NA	0.3	2.1	0.05	0.76	0.05	52.0

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [3] M18/ Road B (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist]				km/h
South: Road B															
1	L2	All MCs	26	0.0	26	0.0	0.081	8.7	LOS A	0.3	1.9	0.44	0.86	0.44	49.7
3	R2	All MCs	29	0.0	29	0.0	0.081	12.7	LOS B	0.3	1.9	0.44	0.86	0.44	49.5
Approach			56	0.0	56	0.0	0.081	10.8	LOS B	0.3	1.9	0.44	0.86	0.44	49.6
East: M18															
4	L2	All MCs	34	0.0	34	0.0	0.103	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
5	T1	All MCs	166	0.0	166	0.0	0.103	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.0
Approach			200	0.0	200	0.0	0.103	1.0	NA	0.0	0.0	0.00	0.10	0.00	58.6
West: M18															
11	T1	All MCs	526	0.0	526	0.0	0.279	0.0	LOS A	0.1	0.8	0.02	0.02	0.02	59.8
12	R2	All MCs	14	0.0	14	0.0	0.279	5.9	LOS A	0.1	0.8	0.02	0.02	0.02	57.0
Approach			540	0.0	540	0.0	0.279	0.2	NA	0.1	0.8	0.02	0.02	0.02	59.7
All Vehicles			796	0.0	796	0.0	0.279	1.1	NA	0.3	1.9	0.05	0.10	0.05	58.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [3.] M18/ Road B** (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Cycles Rate to Depart	Number of Cycles	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh. veh	Dist] m				km/h
South: Road B															
1	L2	All MCs	9	0.0	9	0.0	0.071	9.4	LOS A	0.2	1.6	0.47	0.93	0.47	50.0
3	R2	All MCs	40	0.0	40	0.0	0.071	10.6	LOS B	0.2	1.6	0.47	0.93	0.47	49.8
Approach			49	0.0	49	0.0	0.071	10.3	LOS B	0.2	1.6	0.47	0.93	0.47	49.8
East: M18															
4	L2	All MCs	51	0.0	51	0.0	0.193	5.6	LOS A	0.0	0.0	0.00	0.08	0.00	56.7
5	T1	All MCs	323	0.0	323	0.0	0.193	0.0	LOS A	0.0	0.0	0.00	0.08	0.00	59.2
Approach			374	0.0	374	0.0	0.193	0.8	NA	0.0	0.0	0.00	0.08	0.00	58.8
West: M18															
11	T1	All MCs	181	0.0	181	0.0	0.099	0.1	LOS A	0.1	0.5	0.04	0.05	0.04	59.6
12	R2	All MCs	7	0.0	7	0.0	0.099	6.8	LOS A	0.1	0.5	0.04	0.05	0.04	56.8
Approach			188	0.0	188	0.0	0.099	0.4	NA	0.1	0.5	0.04	0.05	0.04	59.5
All Vehicles			612	0.0	612	0.0	0.193	1.4	NA	0.2	1.6	0.05	0.14	0.05	58.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [4] M18/ Road C (2036 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand		Arrival		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			Flows		Flows					[Veh. Dist]					
			[Total HV] veh/h	%	[Total HV] veh/h	%				v/c	sec				veh
South: Road C															
1	L2	All MCs	4	0.0	4	0.0	0.012	8.7	LOS A	0.0	0.3	0.43	0.82	0.43	49.9
3	R2	All MCs	4	0.0	4	0.0	0.012	12.3	LOS B	0.0	0.3	0.43	0.82	0.43	49.7
Approach			8	0.0	8	0.0	0.012	10.5	LOS B	0.0	0.3	0.43	0.82	0.43	49.8
East: M18															
4	L2	All MCs	13	0.0	13	0.0	0.102	5.6	LOS A	0.0	0.0	0.00	0.04	0.00	57.1
5	T1	All MCs	186	0.0	186	0.0	0.102	0.0	LOS A	0.0	0.0	0.00	0.04	0.00	59.6
Approach			199	0.0	199	0.0	0.102	0.4	NA	0.0	0.0	0.00	0.04	0.00	59.4
West: M18															
11	T1	All MCs	528	0.0	528	0.0	0.276	0.0	LOS A	0.1	0.4	0.01	0.01	0.01	59.9
12	R2	All MCs	7	0.0	7	0.0	0.276	5.7	LOS A	0.1	0.4	0.01	0.01	0.01	57.1
Approach			536	0.0	536	0.0	0.276	0.1	NA	0.1	0.4	0.01	0.01	0.01	59.8
All Vehicles			743	0.0	743	0.0	0.276	0.3	NA	0.1	0.4	0.01	0.03	0.01	59.6

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [4.] M18/ Road C (2036 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	[Total HV]	[Total HV]	[Total HV]				[Veh. veh	Dist] m				km/h
			veh/h	%	veh/h	%	v/c	sec							
South: Road C															
1	L2	All MCs	1	0.0	1	0.0	0.011	9.1	LOS A	0.0	0.3	0.43	0.86	0.43	50.4
3	R2	All MCs	7	0.0	7	0.0	0.011	9.8	LOS A	0.0	0.3	0.43	0.86	0.43	50.1
Approach			8	0.0	8	0.0	0.011	9.7	LOS A	0.0	0.3	0.43	0.86	0.43	50.2
East: M18															
4	L2	All MCs	25	0.0	25	0.0	0.158	5.6	LOS A	0.0	0.0	0.00	0.05	0.00	57.0
5	T1	All MCs	282	0.0	282	0.0	0.158	0.0	LOS A	0.0	0.0	0.00	0.05	0.00	59.5
Approach			307	0.0	307	0.0	0.158	0.5	NA	0.0	0.0	0.00	0.05	0.00	59.3
West: M18															
11	T1	All MCs	166	0.0	166	0.0	0.091	0.1	LOS A	0.1	0.4	0.04	0.05	0.04	59.6
12	R2	All MCs	7	0.0	7	0.0	0.091	6.5	LOS A	0.1	0.4	0.04	0.05	0.04	56.8
Approach			174	0.0	174	0.0	0.091	0.3	NA	0.1	0.4	0.04	0.05	0.04	59.5
All Vehicles			489	0.0	489	0.0	0.158	0.6	NA	0.1	0.4	0.02	0.06	0.02	59.2

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [5] M18/ Glendore Road (2036 + Development)**
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop of Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	101	0.0	101	0.0	0.180	1.0	LOS A	0.9	6.4	0.40	0.49	0.40	55.4
6	R2	All MCs	173	0.0	173	0.0	0.180	6.6	LOS A	0.9	6.4	0.40	0.49	0.40	53.0
Approach			274	0.0	274	0.0	0.180	4.5	NA	0.9	6.4	0.40	0.49	0.40	53.9
North: Glendore Road															
7	L2	All MCs	281	0.0	281	0.0	0.508	10.7	LOS B	3.8	26.3	0.57	0.96	0.78	49.3
9	R2	All MCs	168	0.0	168	0.0	0.508	13.5	LOS B	3.8	26.3	0.57	0.96	0.78	49.0
Approach			449	0.0	449	0.0	0.508	11.8	LOS B	3.8	26.3	0.57	0.96	0.78	49.2
West: M18															
10	L2	All MCs	48	0.0	48	0.0	0.153	5.6	LOS A	0.0	0.0	0.00	0.10	0.00	56.6
11	T1	All MCs	248	0.0	248	0.0	0.153	0.0	LOS A	0.0	0.0	0.00	0.10	0.00	59.0
Approach			297	0.0	297	0.0	0.153	0.9	NA	0.0	0.0	0.00	0.10	0.00	58.6
All Vehicles			1020	0.0	1020	0.0	0.508	6.7	NA	3.8	26.3	0.36	0.58	0.45	52.9

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [5.] M18/ Glendore Road (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%	v/c	sec		[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	114	0.0	114	0.0	0.190	0.6	LOS A	1.0	6.9	0.31	0.43	0.31	55.7
6	R2	All MCs	198	0.0	198	0.0	0.190	6.1	LOS A	1.0	6.9	0.31	0.43	0.31	53.3
Approach			312	0.0	312	0.0	0.190	4.1	NA	1.0	6.9	0.31	0.43	0.31	54.1
North: Glendore Road															
7	L2	All MCs	92	0.0	92	0.0	0.138	8.4	LOS A	0.5	3.8	0.26	0.87	0.26	50.7
9	R2	All MCs	51	0.0	51	0.0	0.138	10.2	LOS B	0.5	3.8	0.26	0.87	0.26	50.5
Approach			142	0.0	142	0.0	0.138	9.0	LOS A	0.5	3.8	0.26	0.87	0.26	50.6
West: M18															
10	L2	All MCs	103	0.0	103	0.0	0.098	5.6	LOS A	0.0	0.0	0.00	0.33	0.00	54.8
11	T1	All MCs	83	0.0	83	0.0	0.098	0.0	LOS A	0.0	0.0	0.00	0.33	0.00	57.1
Approach			186	0.0	186	0.0	0.098	3.1	NA	0.0	0.0	0.00	0.33	0.00	55.8
All Vehicles			640	0.0	640	0.0	0.190	4.9	NA	1.0	6.9	0.21	0.50	0.21	53.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY



Site: [6] M18/ Proposed Access (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing AM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%				[Veh. veh	Dist]				
			veh/h		veh/h		v/c	sec		veh	m				km/h
East: M18															
5	T1	All MCs	211	0.0	211	0.0	0.250	2.0	LOS A	1.3	8.9	0.48	0.56	0.48	55.8
6	R2	All MCs	134	0.0	134	0.0	0.250	8.4	LOS A	1.3	8.9	0.48	0.56	0.48	53.4
Approach			344	0.0	344	0.0	0.250	4.5	NA	1.3	8.9	0.48	0.56	0.48	54.8
North: Proposed Access															
7	L2	All MCs	389	0.0	389	0.0	0.940	29.4	LOS D	14.6	102.5	0.95	1.79	3.40	39.1
9	R2	All MCs	167	0.0	167	0.0	0.940	37.7	LOS E	14.6	102.5	0.95	1.79	3.40	38.9
Approach			557	0.0	557	0.0	0.940	31.9	LOS D	14.6	102.5	0.95	1.79	3.40	39.0
West: M18															
10	L2	All MCs	58	0.0	58	0.0	0.308	5.6	LOS A	0.0	0.0	0.00	0.06	0.00	56.9
11	T1	All MCs	539	0.0	539	0.0	0.308	0.1	LOS A	0.0	0.0	0.00	0.06	0.00	59.3
Approach			597	0.0	597	0.0	0.308	0.6	NA	0.0	0.0	0.00	0.06	0.00	59.1
All Vehicles			1498	0.0	1498	0.0	0.940	13.1	NA	14.6	102.5	0.46	0.82	1.37	48.8

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.

MOVEMENT SUMMARY

 **Site: [6.] M18/ Proposed Access** (2036 + Development)
Output produced by SIDRA INTERSECTION Version: 10.0.5.217

ERF 11305 WALMER TOWNSHIP
TRAFFIC IMPACT ASSESSMENT
2036 Existing PM + Development
Site Category: Existing Design
Stop (Two-Way)
Site Scenario: 1 | Local Volumes

Vehicle Movement Performance															
Mov ID	Turn	Mov Class	Demand Flows		Arrival Flows		Deg. Satn	Aver. Delay	Level of Service	95% Back Of Queue		Prop. Qued	Eff. Stop Rate	Number of Cycles to Depart	Aver. Speed
			[Total HV]	%	[Total HV]	%				[Veh. veh	Dist] m				km/h
East: M18															
5	T1	All MCs	327	0.0	327	0.0	0.452	1.8	LOS A	3.5	24.3	0.49	0.56	0.57	55.4
6	R2	All MCs	363	0.0	363	0.0	0.452	7.6	LOS A	3.5	24.3	0.49	0.56	0.57	53.0
Approach			691	0.0	691	0.0	0.452	4.9	NA	3.5	24.3	0.49	0.56	0.57	54.1
North: Proposed Access															
7	L2	All MCs	156	0.0	156	0.0	0.324	9.2	LOS A	1.5	10.4	0.53	0.85	0.57	48.9
9	R2	All MCs	66	0.0	66	0.0	0.324	19.2	LOS C	1.5	10.4	0.53	0.85	0.57	48.7
Approach			222	0.0	222	0.0	0.324	12.2	LOS B	1.5	10.4	0.53	0.85	0.57	48.9
West: M18															
10	L2	All MCs	156	0.0	156	0.0	0.181	5.6	LOS A	0.0	0.0	0.00	0.27	0.00	55.2
11	T1	All MCs	188	0.0	188	0.0	0.181	0.0	LOS A	0.0	0.0	0.00	0.27	0.00	57.5
Approach			344	0.0	344	0.0	0.181	2.6	NA	0.0	0.0	0.00	0.27	0.00	56.5
All Vehicles			1257	0.0	1257	0.0	0.452	5.5	NA	3.5	24.3	0.36	0.53	0.42	53.7

Site Level of Service (LOS) Method: Delay (SIDRA). Site LOS Method is specified in the Parameter Settings dialog (Options tab).
Vehicle movement LOS values are based on average delay per movement.
Minor Road Approach LOS values are based on average delay for all vehicle movements.
NA (TWSC): Level of Service is not defined for major road approaches or the intersection as a whole for Two-Way Sign Control (HCM LOS rule).
Two-Way Sign Control Capacity Model: SIDRA Standard.
Delay Model: SIDRA Standard (Control Delay: Geometric Delay is included).
Queue Model: SIDRA queue estimation methods are used for Back of Queue and Queue at Start of Gap.
Gap-Acceptance Capacity Formula: SIDRA Standard (Akçelik M3D).
HV (%) values are calculated for All Movement Classes of All Heavy Vehicle Model Designation.
Arrival Flows used in performance calculations are adjusted to include any Initial Queued Demand and Upstream Capacity Constraint effects.